The “Depression Trap”: Genetic Risk, Labor Supply, and the Case for Precision Psychiatry

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Mental ill health is common, and costly through work

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up from 17.6% in 2007

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2.8 million

out of work through long-term sickness, near a record high

over half report depression or anxiety

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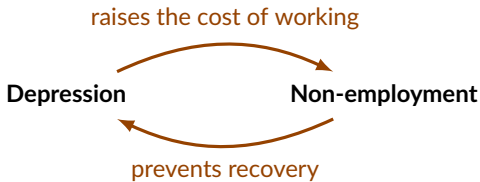
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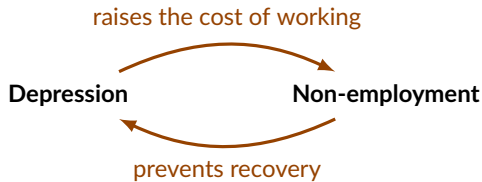
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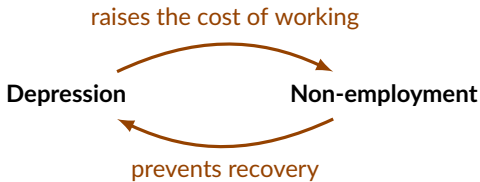


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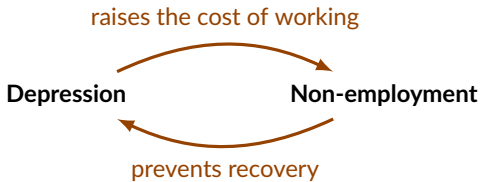
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RQ3 Can targeting therapy by genetic risk, and expanding it, cost-effectively improve mental health and work?

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Molecular genetic data linked to a UK panel (men 25–55, 1991–2023); event studies of depression recurrences and redundancies; an estimated life-cycle model of work, mental health and therapy.

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RQ3 **Targeting and expanding therapy both pay for themselves.** Priority for above-median risk raises clinical efficacy by 29% and funds universal genotyping; 10% more therapists self-finance.

What's new? Position in the literature

We document the persistence of the depression trap, use genetics to predict who enters it, and provide the first structural evaluation of targeting and expanding mental health care.

- **Economics of mental health**

  Mental health and work: Ettner et al., 1997; Frijters et al., 2014; Bryan et al., 2022; Biasi et al., 2021
  Job loss and mental health: Marcus, 2013; Picchio and Ubaldi, 2024; Sterud et al., 2025
  RCTs: work $\rightarrow$ mental health (Hussam et al., 2022); therapy $\rightarrow$ work (Smith et al., 2025)

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- **Genoeconomics**

  Reviews: Biroli et al., 2022; Benjamin et al., 2024
  Polygenic indices and economic outcomes: Papageorge and Thom, 2020; Barth et al., 2020
  Genetics in structural models: Barth et al., 2022; Houmark et al., 2024; Bolt et al., 2025; Agostinelli and Weingarten, 2025; Yang et al., 2026
  Depression PGI: education (Amin et al., 2019); labor market (Martin Bassols et al., 2024)

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- **Structural models of health and labor supply**

  Physical health: French, 2005; Hosseini et al., 2021; De Nardi et al., 2024; Capatina and Keane, 2026
  Mental health and work: Jolivet and Postel-Vinay, 2025; Diaz-Campo et al., 2025
  Mental health treatment: Cronin et al., 2024; Abramson et al., 2024; Sauer et al., 2025

  $\Rightarrow$ **first structural evaluation of the fiscal case for targeting and expanding therapy**

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- **Public economics of health care**

  Tagging and prediction: Akerlof, 1978; Logue and Slemrod, 2008; Kleinberg et al., 2015
  Rationing by waiting: Gravelle and Siciliani, 2009; Aakvik et al., 2015; Godøy et al., 2024
  Fiscal returns: Layard et al., 2007; Hendren and Sprung-Keyser, 2020; Serena, 2025

  $\Rightarrow$ the value of a genetic tag for rationed therapy, beyond observed histories

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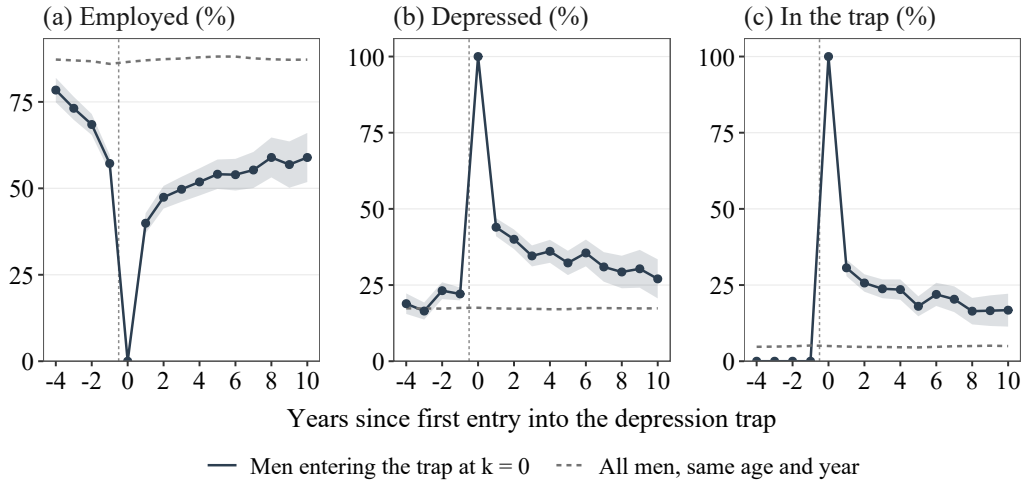
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- **Sample:** men 25–55 in paid employment, unemployment or long-term sickness: ≈29,000 men, ≈160,000 person-waves

- **Genetics:** ≈10,000 participants genotyped; ≈2,200 men (≈21,000 person-waves) after the same restrictions

▸ GHQ-12 ▸ Details ▸ Women

Men who enter the depression trap stay out of work, and depressed



Why? Depression → non-employment, non-employment → depression, or selection.

Does depression lower employment? The timing of a recurrence

Why recurrence. A first episode is often triggered by a major life event; later episodes need less and less of a trigger (Post, 1992; Kendler et al., 1999) and are hard to predict in advance (Moriarty et al., 2021). ▸ Literature

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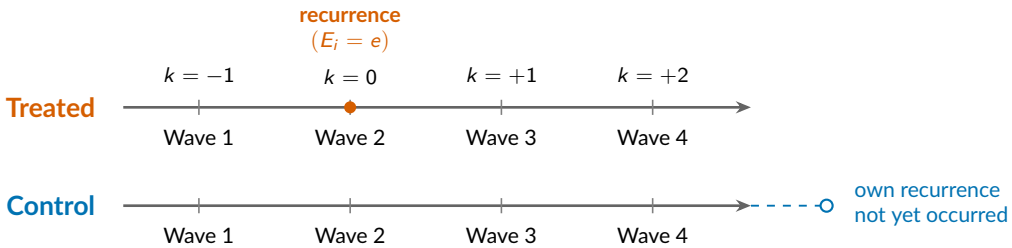
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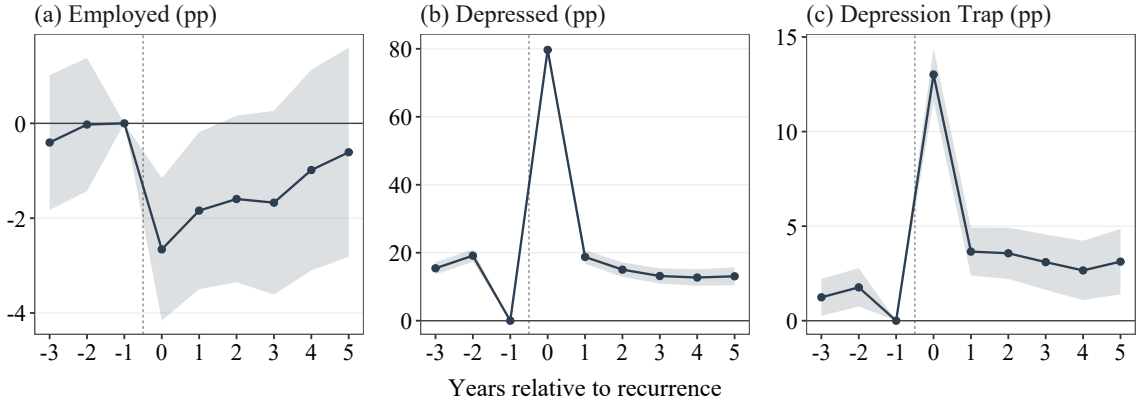
Specification. Sun and Abraham (2021) event study, men who recur vs. those who have not (yet):

$$Y_{it} = \gamma_t + \sum_e \sum_{k \neq -1} \delta_{e,k} \, \mathbf{1}\{E_i = e\} \, \mathbf{1}\{t - E_i = k\} + \mathbf{X}'_{it}\beta + \varepsilon_{it}$$

E_i: wave of the recurrence; γ_t: wave effects; $\mathbf{X}_{it}$: age, age^2.



Does depression lower employment? The timing of a recurrence



A recurrence lowers employment by 2.7 pp

Trap $+3$ pp for five years; literature (employment): 1.5–9 pp

▸ Robustness ▸ Literature ▸ Women ▸ By later work ▸ Mechanisms

Does job loss worsen mental health? Redundancy as a firm-side shock

Why redundancy. In UK law the *position* is eliminated and the worker is not at fault: firm-side job loss, as in the plant-closure literature (Picchio and Ubaldi, 2024). ▸ Literature

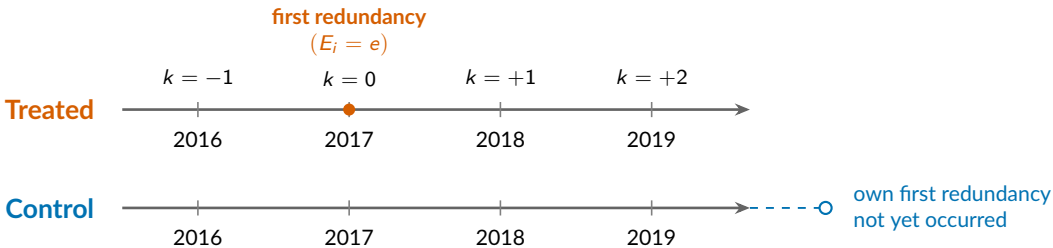
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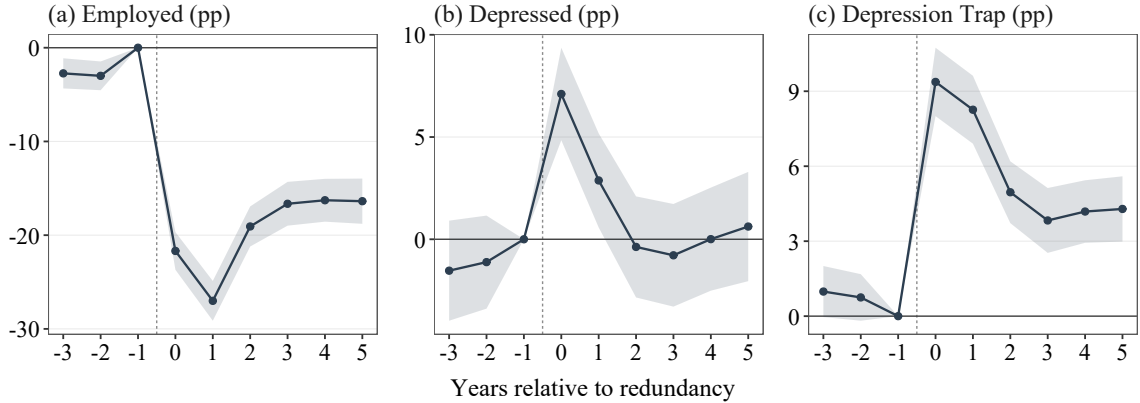
Specification. Same estimator, redundant vs. not-yet-redundant men:

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E_i: year of first redundancy; γ_t: interview-year effects; $\mathbf{X}_{it}$: age, age^2, education FE.



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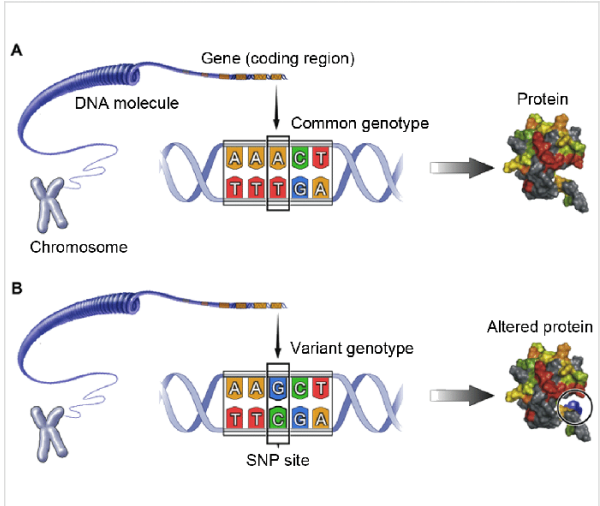


A redundancy increases depression by 7 pp

Symptoms $+0.22$ s.d.; literature: 0.10–0.27 s.d.

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Measuring genetic risk before the fact: genetic heterogeneity



Measuring genetic risk before the fact: the polygenic index

- A PGI is a weighted genetic index: for individual i,

$$\mathrm{PGI}_i = \sum_{j=1}^{J} \hat{\beta}_j \, G_{ij},$$

where $G_{ij} \in \{0, 1, 2\}$ counts copies of allele j and the weights $\hat{\beta}_j$ come from an external GWAS.

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▸ Details

We use the PGI for depressive symptoms (Baselmans et al., 2019)

- **Predictive:** +1 s.d. ⇒ +6.9 pp chance of depression ▸ Prediction
- **Preventive:** most depression is treated late, or never
  - only 8–9% in England use NHS mental-health services each year [NHS England, 2024]
  - over half with depression or anxiety get no treatment [Morris et al., 2025]
  - 8 years from onset to first treatment (US median) [Wang et al., 2005]
- **Cheap:** genetic test US$20 vs. therapy course £1,000 [Raben et al., 2022; NHS England, 2023a,c]

Genetic risk increases entry into the trap and slows exit

Annual transition probability (%)	Below-median risk	Above-median risk	Difference
Mental health			
Onset (healthy → depressed)	9.3	12.8	$+3.5^{***}$
Recovery (depressed → healthy)	64.4	49.1	-15.3^{***}

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Employment			
Exit (employed → non-employed)	1.6	2.3	$+0.7^{**}$
Entry (non-employed → employed)	38.8	25.1	-13.7^{***}

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"Depression trap" (non-employed & depressed)			
Entry into trap	1.0	1.7	$+0.7^{***}$
Exit from trap	60.2	35.8	-24.4^{***}

Entry 70% higher and exit 40% lower: expected spells of 2.8 against 1.7 years.

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The same shocks cost high-risk men more work

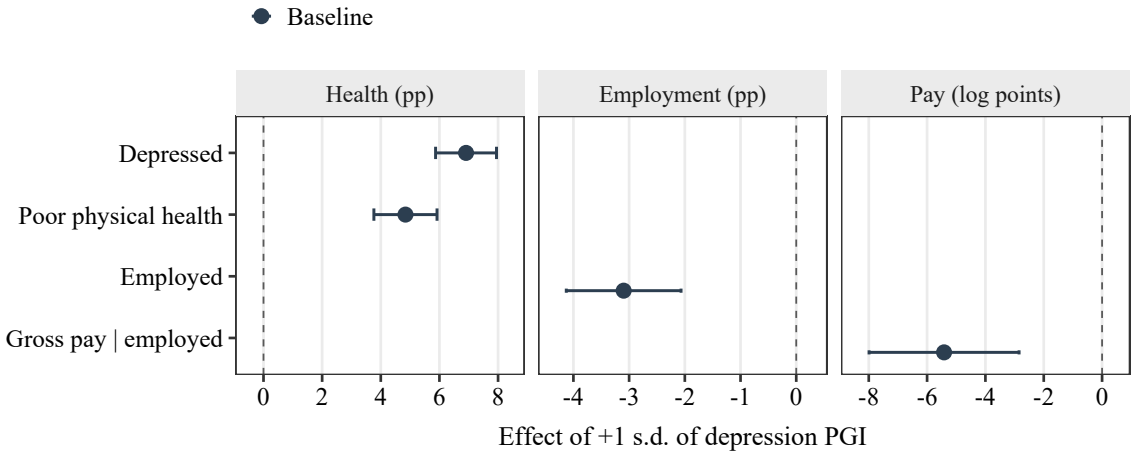
Effect at impact ($k = 0$), pp	High risk	Low risk	Difference
Recurrence of a depressive episode			
Employed	-7.5	-0.9	-6.6^{**}
Depressed	$+83.1$	$+87.4$	-4.2^{***}
In the trap	$+15.4$	$+7.3$	$+8.1^{***}$
Redundancy			
Employed	-28.7	-16.5	-12.3^{**}
Depressed	$+10.8$	$+4.0$	$+6.8$
In the trap	$+13.6$	$+6.5$	$+7.1^{**}$

In both designs, employment and the trap respond more at high genetic risk; depression does not respond significantly more.

Genotyped men split in two by the depression PGI; event studies as above, by group. ** $p < 0.05$, *** $p < 0.01$.

▸ Recurrence G×E ▸ Redundancy G×E

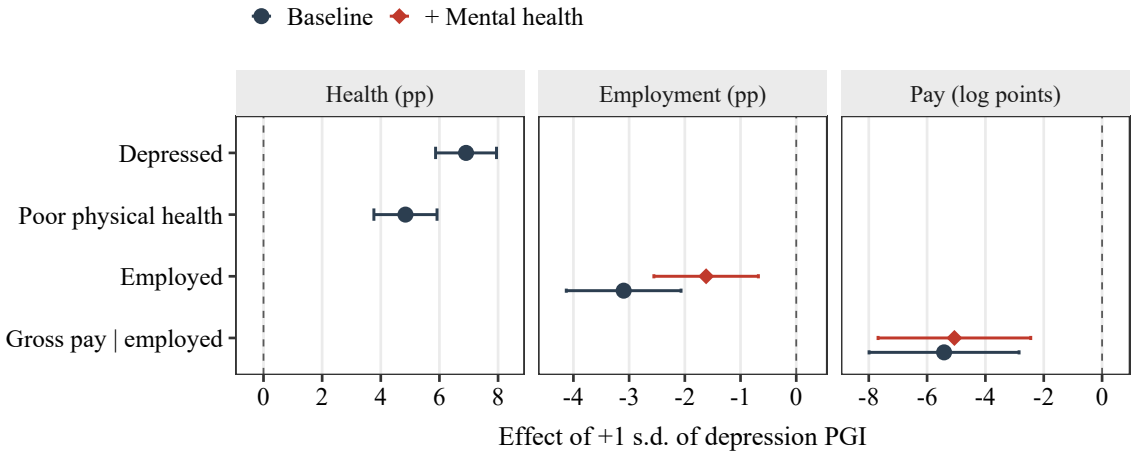
Employment falls mainly through mental and physical health



Employment gap of 3.1 pp per s.d.

▸ Within-Family ▸ Women

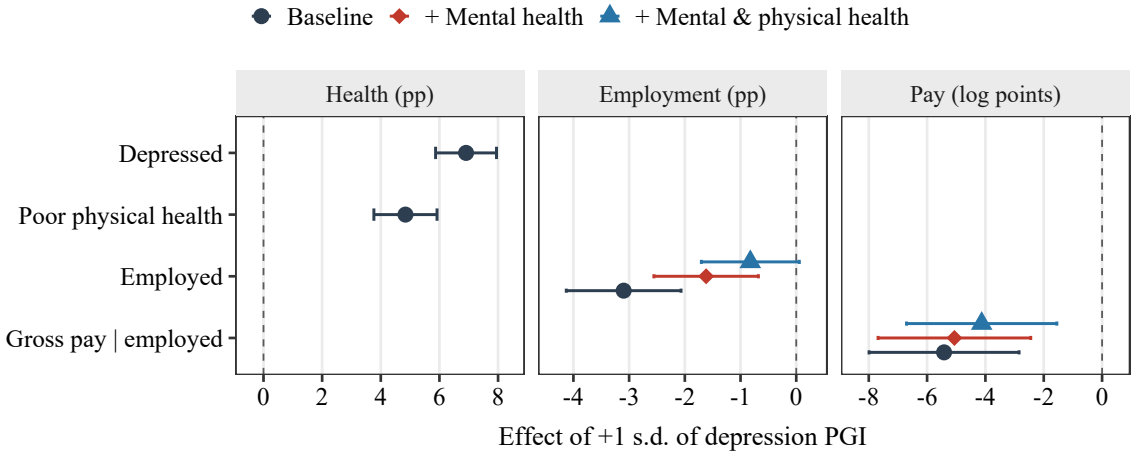
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Employment gap of 3.1 pp per s.d.: 48% through mental health

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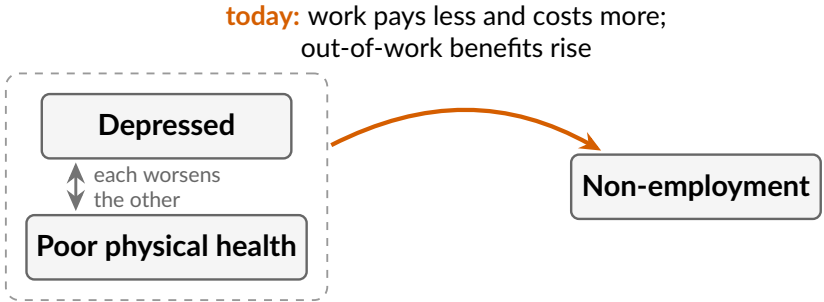
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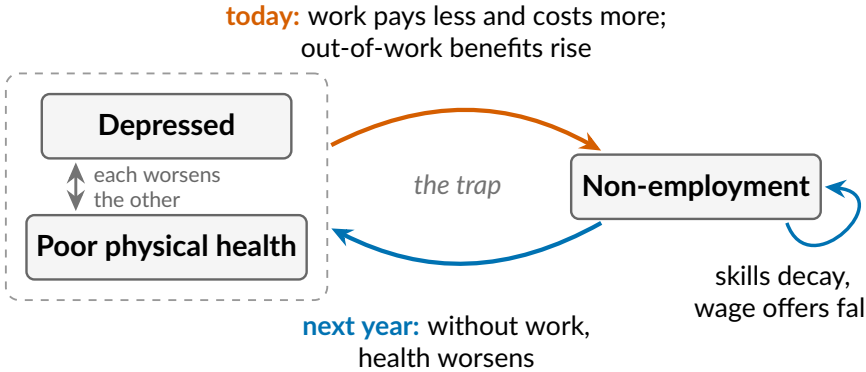
Employment gap of 3.1 pp per s.d.: 48% through mental health, 73% with physical health

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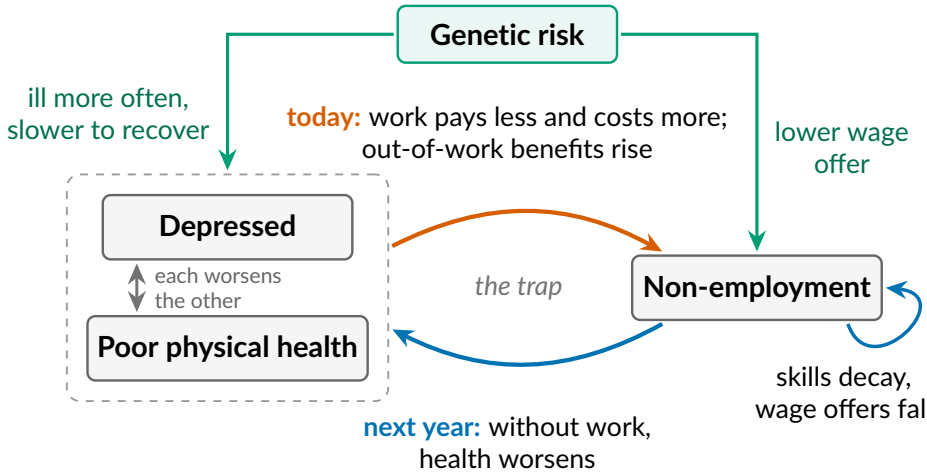
The mechanism: depression and non-employment feed each other



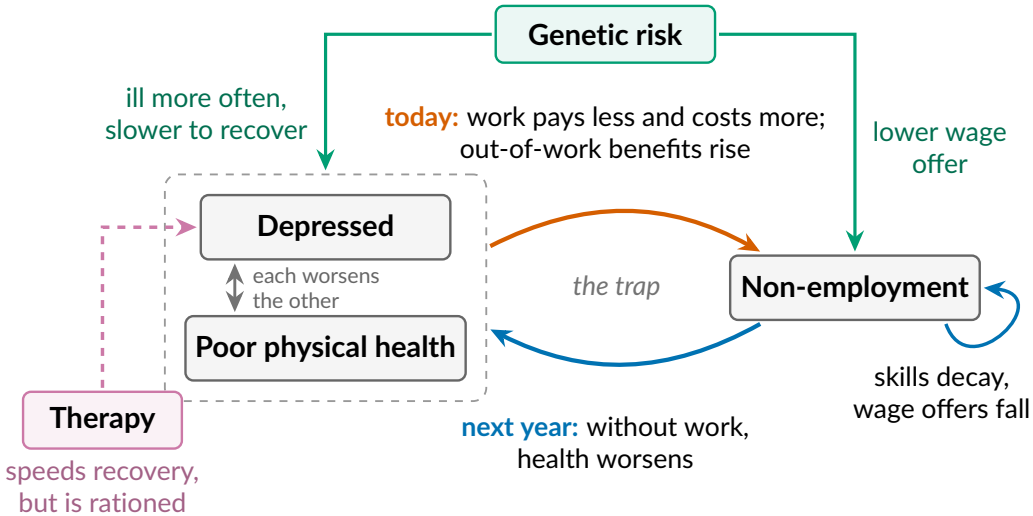
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Model overview

$$\underbrace{\text{Value today}}_{\substack{\text{State variables today}\\ \{\text{age, savings, skills, wage shock, mental health, physical health, genetic risk, type}\}}} = \max_{\text{choices}} \Big\{ \text{Payoff today} + \beta \mathbb{E}\Big[\underbrace{\text{Value tomorrow}}_{\substack{\text{Endogenous states}\\ \{\text{savings, skills, mental health, physical health}\}}\ \underbrace{}_{\substack{\text{Exogenous states}\\ \{\text{age, wage shock, genetic risk, type}\}}} } \Big] \Big\}$$

$\underbrace{\{\text{apply for therapy, work, save}\}}_{\text{Choice variables}}$

▸ Timing ▸ Retirement ▸ Full model

Payoff today: poor health makes work cost more

$$\text{Value today} = \max_{\text{choices}} \left\{ \text{Payoff today} + \beta\, \mathbb{E}\left[\text{Value tomorrow} \right] \right\}$$

$$\left\{ \begin{array}{ll} \text{age} & \text{mental health} \\ \text{savings} & \text{physical health} \\ \text{skills} & \text{genetic risk} \\ \text{wage shock} & \text{type} \end{array} \right\} \quad \left(\begin{array}{l} \text{apply for therapy} \\ \text{work} \\ \text{save} \end{array} \right) \quad \left\{ \begin{array}{l} \text{savings} \\ \text{skills} \\ \text{mental health} \\ \text{physical health} \end{array} \right\} \left\{ \begin{array}{l} \text{age} \\ \text{wage shock} \\ \text{genetic risk} \\ \text{type} \end{array} \right\}$$

c consumption L leisure n work τ therapy received a, a' savings now, next year mh, ph mental, physical health

$$u = \frac{\left(c^{\gamma} L^{1-\gamma}\right)^{1-\omega}}{1-\omega} - \underbrace{\psi(n)\,\tau}_{\substack{\text{therapy's time and stigma} \\ \text{cost, higher in work}}}$$

$$L = 1 - \underbrace{\ell_n\, n}_{\text{hours of work}} - \underbrace{\delta(mh, ph)}_{\text{time lost to poor health}}$$

$$c + a' = (1+r)\,a + \underbrace{n\,w\,[1-\mathcal{T}(nw)]}_{\text{earnings after tax}} + \underbrace{b(n, mh, ph)}_{\text{benefits, higher in poor health}} - \underbrace{P_{\tau}(n)\,\tau}_{\substack{\text{therapy price,} \\ \text{zero out of work}}}, \qquad a' \geq 0$$

$\Rightarrow$ Poor health incurs a time cost, making working more costly; benefits make not working pay more.

▸ Full model

Wage offers: out of work, skills decay and offers fall

$$\text{Value today} = \max_{\text{choices}} \left\{ \text{Payoff today} + \beta\, \mathbb{E}\left[\text{Value tomorrow} \right] \right\}$$

$$\left\{ \begin{array}{ll} \text{age} & \text{mental health} \\ \text{savings} & \text{physical health} \\ \text{skills} & \text{genetic risk} \\ \text{wage shock} & \text{type} \end{array} \right\} \quad \left(\begin{array}{l} \text{apply for therapy} \\ \text{work} \\ \text{save} \end{array} \right) \quad \left\{ \begin{array}{l} \text{savings} \\ \text{skills} \\ \text{mental health} \\ \text{physical health} \end{array} \right\} \left\{ \begin{array}{l} \text{age} \\ \text{wage shock} \\ \text{genetic risk} \\ \text{type} \end{array} \right\}$$

w wage offer $\quad hc,\ hc'$ skills now, next year $\quad g$ genetic risk $\quad x$ type $\quad \epsilon$ wage shock $\quad n$ work

$$\ln w = \underbrace{\beta_1\, g}_{\text{genetic risk}} + \underbrace{\beta_2\, x}_{\text{type}} + \underbrace{\beta_3'\, mh + \beta_4'\, ph}_{\text{health}} + \underbrace{f(hc)}_{\text{skills}} + \underbrace{\epsilon}_{\substack{\text{persistent} \\ \text{shock}}}$$

$$hc' = \underbrace{(1 - \delta_{hc})\, hc}_{\text{skills decay}} + \underbrace{n}_{\text{and grow in work}}$$

$\Rightarrow$ The longer a spell out of work, the lower the wage offer that would end it.

▸ Full model

Health next year: work and therapy both improve it

$$\text{Value today} = \max_{\text{choices}} \left\{ \text{Payoff today} + \beta\, \mathbb{E}\Big[\text{Value tomorrow} \Big] \right\}$$

$$\begin{Bmatrix} \text{age} \\ \text{savings} \\ \text{skills} \\ \text{wage shock} \end{Bmatrix} \begin{matrix} \text{mental health} \\ \text{physical health} \\ \text{genetic risk} \\ \text{type} \end{matrix} \qquad \begin{pmatrix} \text{apply for therapy} \\ \text{work} \\ \text{save} \end{pmatrix} \qquad \begin{Bmatrix} \text{savings} \\ \text{skills} \\ \text{mental health} \\ \text{physical health} \end{Bmatrix} \begin{Bmatrix} \text{age} \\ \text{wage shock} \\ \text{genetic risk} \\ \text{type} \end{Bmatrix}$$

mh, ph good, moderate or severe $mh^{*\prime}$ index of next year's mental health τ therapy received n work

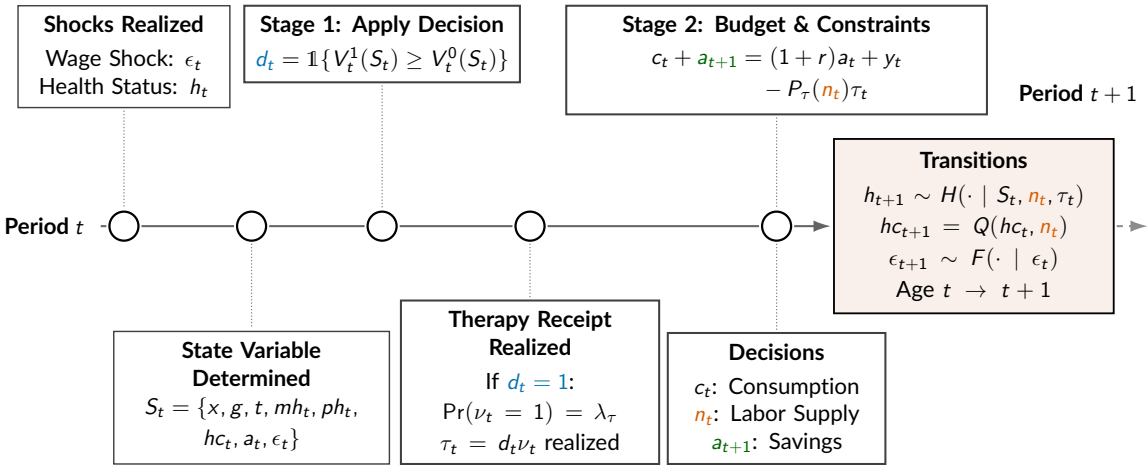
$$mh^{*\prime} = \underbrace{\mu_1' \, mh}_{\text{persistence}} + \underbrace{\mu_2 \, \tau}_{\text{therapy}} + \underbrace{\mu_3 \, n}_{\text{work}} + \underbrace{\mu_4' \, ph}_{\text{physical health}} + \underbrace{\mu_5'(g, x, \text{age})}_{\text{genetic risk, type, age}} + \upsilon'$$

- Ordered logit, two cut-offs; physical health: same form, no therapy
- $\tau = 1$: applies *and* offered (probability λ_τ)
- Therapy effect μ_2: calibrated from RCTs

$\Rightarrow$ Out of work: worse mental health. Therapy: better, if offered.

▸ Health ▸ Full model

Timing within a period



▸ Retirement

Estimation and identification

1. Types

k-means on wage and health, net of age and genetics → five permanent types (Bonhomme et al., 2022)

↓

2. Outside the model

physical-health transitions, pensions, benefits, taxes; therapy effect: RCTs; access: NHS data

↓

3. Inside the model: SMM

mental-health transitions, wages, preferences: jointly, given 1–2

▸ Types ▸ Therapy effect ▸ Details

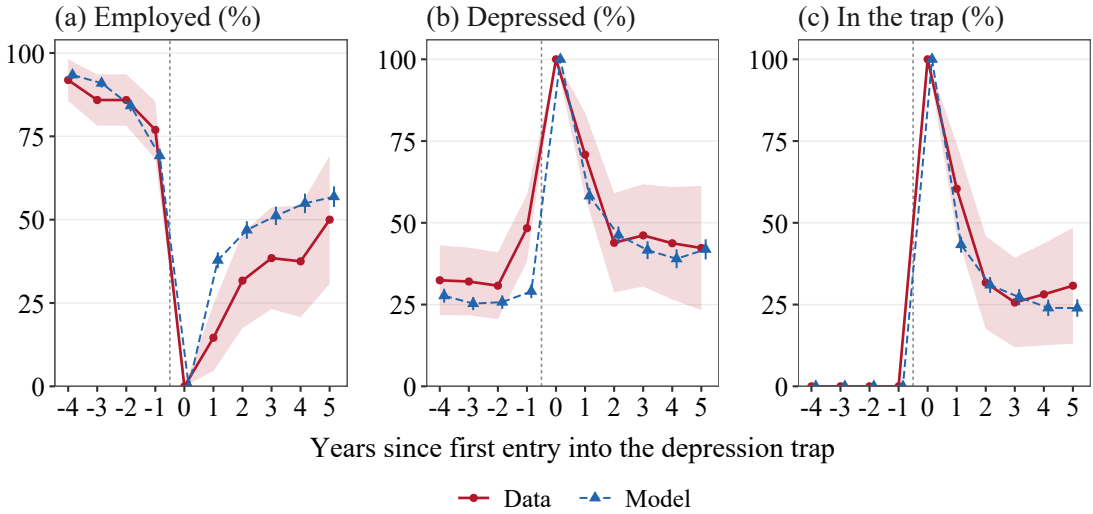
Inside the model (SMM): targeted moments

Parameter	Description	Targeted moments
$\beta_0, \ldots, \beta_6$	Wage offer process	Mean wages: age, health, type, genes, work history
δ_{hc}	Human capital depreciation	Age at the peak of the wage profile
ρ, σ_η	Persistent wage shocks	Wage dispersion: health, type, genes, age
μ_1, μ_3–μ_5	Mental health dynamics	Transitions: health, work, therapy, age, type, genes
$\delta(mh_t, ph_t)$	Time cost of poor health	Employment by MH and PH
$\psi(n_t)$	Non-pecuniary therapy cost	Therapy take-up by employment status
γ	Consumption share	Labor force participation
β	Discount factor	Aggregate saving ratio

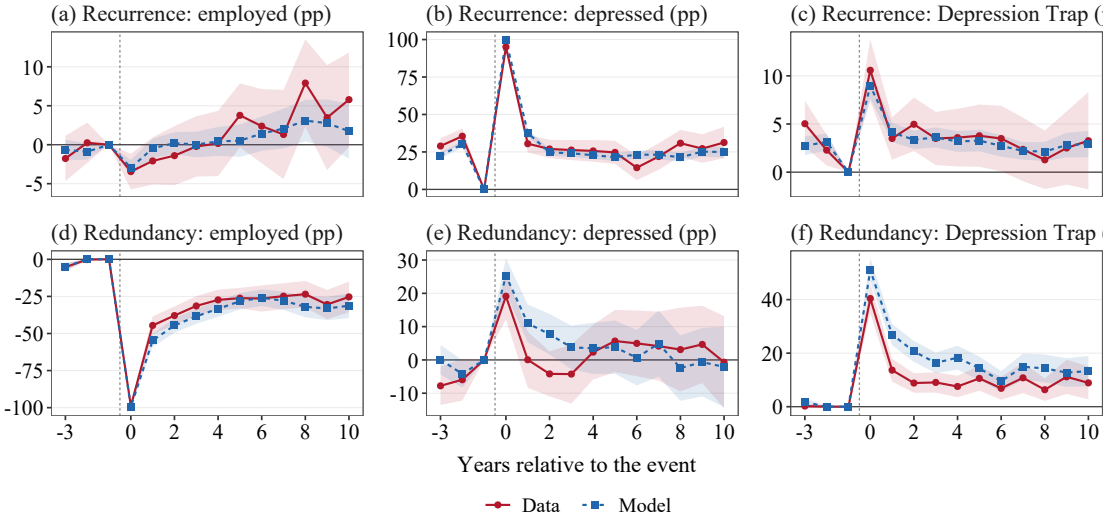
Joint SMM estimation; most informative moments shown. μ_2 (therapy): RCTs.

▸ Health fit ▸ Earnings fit ▸ Employment, therapy, saving ▸ Event studies

Untargeted Model Fit: first entry into the trap

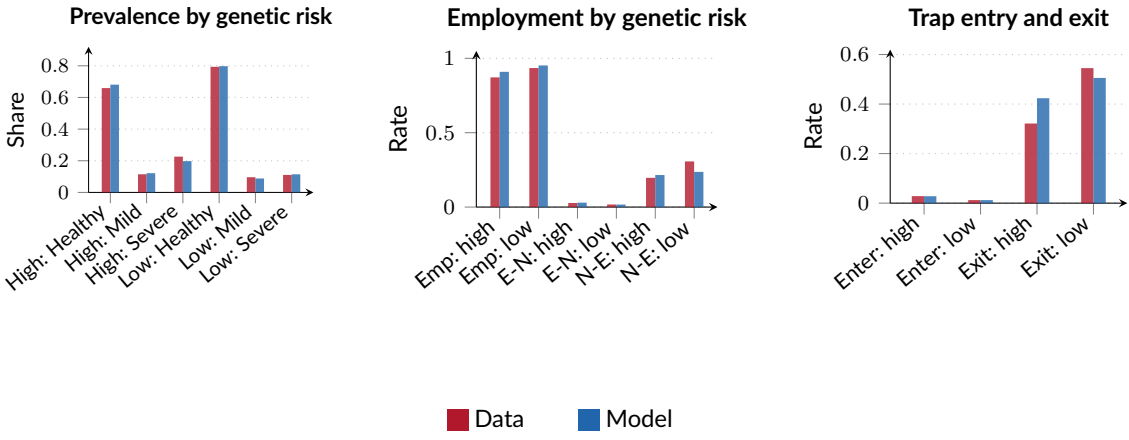


Untargeted Model Fit: event studies



▸ Recurrence ▸ Redundancy

Untargeted Model Fit: gradients by genetic risk



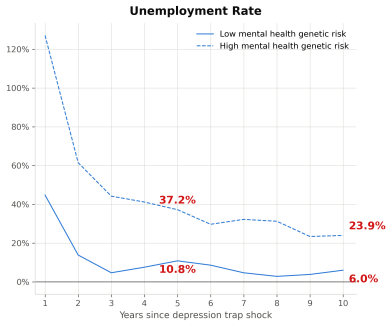
▸ Recurrence by risk ▸ Job loss by risk ▸ By age

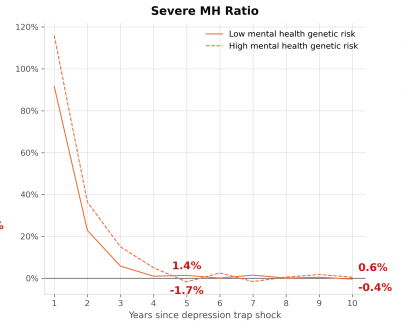
Effects of a one-year trap shock persist for ten years

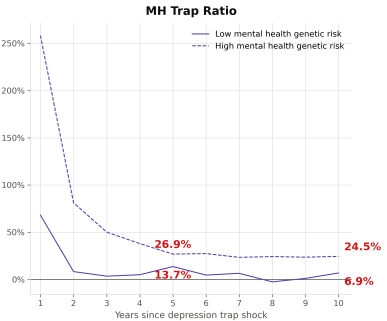
Shock: year 1, everyone in the trap (severe depression, out of work)
Plotted: gap from baseline (%), ten years

Effects of a one-year trap shock persist for ten years

Shock: year 1, everyone in the trap (severe depression, out of work)
Plotted: gap from baseline (%), ten years







Episode over within four years; employment gap persists
Year-10 unemployment: +23.9% high risk vs +6.0% low

▸ Conditioning

Priority access by genetic risk raises the return per course

Benchmark: offer probability $\lambda = 0.69$ for all applicants (NHS data)
Priority X**:** shorter wait at high genetic risk; total courses fixed

$$\underbrace{\frac{1-\lambda(H)}{\lambda(H)}}_{\substack{\text{expected wait (years),}\\ \text{above-median risk}}} = (1-X) \underbrace{\frac{1-\lambda(L)}{\lambda(L)}}_{\substack{\text{expected wait (years),}\\ \text{below-median risk}}}$$

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Example: $X = 75\%$**.**

	Treated (%), by risk		Fiscal effect (£ per person)			Severe-depression
Priority X	Below	Above	Tax	Benefits	Net	years averted per course
Benchmark	1.22	2.51	—	—	—	0.415
75%	0.45	3.33	8.5	16.8	25.3	0.534

+29% severe-depression years averted per course
Net fiscal surplus: £335 per course reallocated (five reallocated fund two more)

▸ Implementation ▸ All priorities ▸ Who applies

Genetic information pays for itself

No genotype: infer risk from health, work and earnings histories
Survey accuracy: AUC ≈ 0.65

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		Fiscal effect (£ per person)		Years of severe
Priority X	Information	Tax + benefits	Net	depression averted per course
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	Histories only	2.7 + 4.6	7.4	0.437

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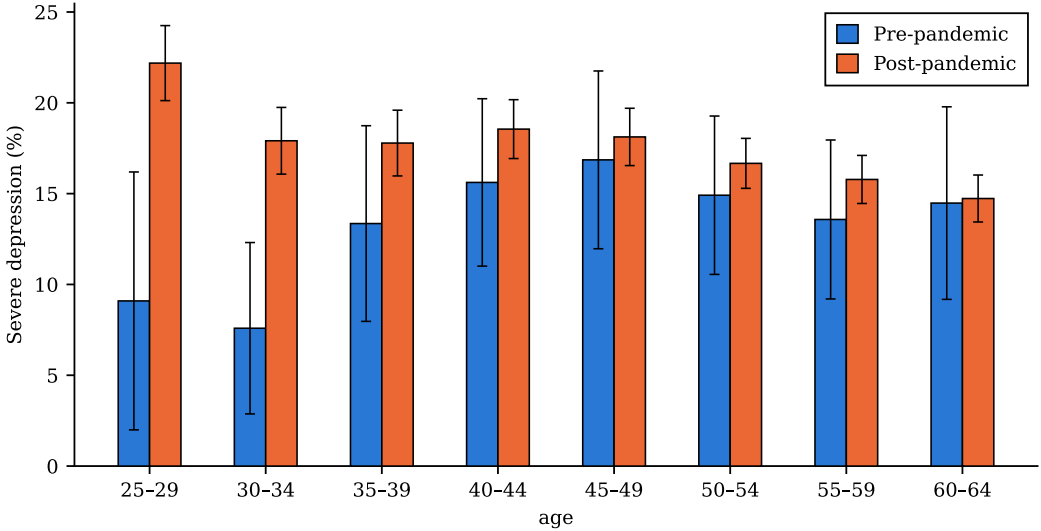
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Genotype: $+$**£18 per person** at $X = 75\%$, above the US$20 cost of genotyping
$\Rightarrow$ Better targeting alone pays for genotyping

All priorities

Severe depression has risen most among the young

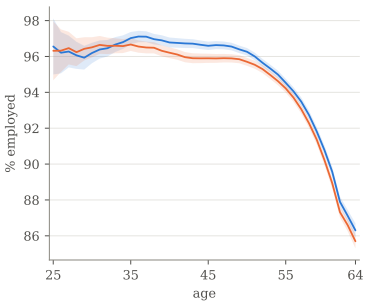


The mental-health shift propagates over the working life

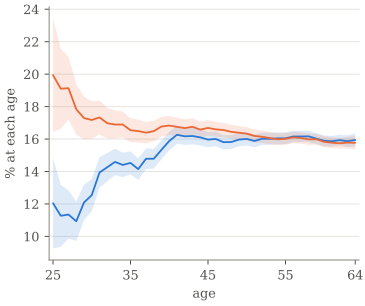
$\rightarrow$ Pre-pandemic population; only the age profile of mental health is moved to match the post-pandemic data. Genotype, type, physical health, assets and human capital fixed.

The mental-health shift propagates over the working life

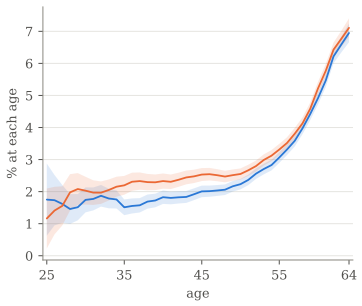
→ Pre-pandemic population; only the age profile of mental health is moved to match the post-pandemic data. Genotype, type, physical health, assets and human capital fixed.



(a) Employment rate



(b) Severe depression



(c) Depression trap

Pre-pandemic Post-pandemic

Fiscal cost: £804 per person in present value at 25 (tax −£529, benefits +£275).

A 10% larger therapist workforce pays for itself

NHS plan: $+14\%$ courses by 2028/29; £600m for 15,000 therapists (NHS England, 2025, 2023b)
Here: offer probability $+10\%$

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Here: offer probability $+10\%$

	Fiscal effect (£1,000 per additional therapist)				Severe-depression
Workforce	Tax	Benefit savings	Therapist wages	Net	person-years averted
$+10\%$	226	146	316	$+56$	482

Net $+$£56k per therapist after wages, vs £40k training cost
482 person-years of severe depression averted
Self-financing (excl. training) up to $\approx$14% expansion

▸ All expansions ▸ More counterfactuals

Conclusions

1. **Evidence consistent with a self-reinforcing cycle**
   - In line with the literature: a recurrence lowers employment by 2.7 pp; a redundancy raises depression by 7 pp
   - A one-year spell in the trap leaves unemployment 24% above baseline ten years on at high genetic risk, 6% at low

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1. **Evidence consistent with a self-reinforcing cycle**
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2. **Genetic risk identifies who is at risk before the fact**
   - 70% more entry, 40% slower exit; predicts as well as symptoms today, and for longer
   - The same shocks cost high-risk men more work, not more symptoms
3. **Precision psychiatry: targeting therapy on genetic risk is self-financing**
   - Priority by genetic risk: 29% more severe-depression years averted per course, £25 per person to the exchequer
   - The gain needs the genotype; a 10% larger therapist workforce also self-finances

Thank you

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Polygenic index: the genome-wide association study

- In a *discovery sample* (UK Biobank, 23andMe), for each SNP $j \in \{1, \ldots, J\}$ regress the outcome on the genotype, one SNP at a time:

$$y_i = \alpha_j + \beta_j G_{ij} + \epsilon_{ij}, \qquad G_{ij} \in \{0, 1, 2\}$$

- Adjust the estimates $\{\hat{\beta}_j\}$ for statistical issues (e.g. correlation between neighbouring SNPs) to get the weights $\{\beta_j^*\}$.

- Apply the weights to our men: weights from an independent sample (Baselmans et al., 2019); ancestry controls; estimates unchanged when comparing siblings.

Within-family Back

Job loss and mental health: the literature

Evidence	Effect on depressive symptoms	Source
Meta-analysis, 38 longitudinal studies, prior mental health controlled	job loss $+0.19$ s.d.; re-employment -0.27	[Sterud et al., 2025]
Meta-analysis, 65 economics studies	job loss $\approx +0.10$ s.d. (men, mental health)	[Picchio and Ubaldi, 2024]
Plant closures, Germany	job loss $+0.27$ s.d.; spouse $+0.18$ s.d.	[Marcus, 2013]
RCT: paid work vs. equal cash	work -0.24 s.d.; cash ≈ 0	[Hussam et al., 2022]
Our panel, individual FE: in work at $t-1$	depression onset -3.4 pp, recovery $+3.6$ pp	▸ Fixed effects
Our redundancy event study	$+0.22$ s.d.	

For scale: antidepressants 0.30 s.d., psychotherapy 0.60 s.d. [Cipriani et al., 2018; Hofmann et al., 2012]

◂ Back

Depression and employment: the literature

Evidence	As reported	Scaled to 1.5 s.d.	Source
RCT of talking therapy (Norway)	symptoms -0.88 s.d.; in work $+5$ pp in years 2–5	≈ -9 pp	[Smith et al., 2025]
Talking therapy, waiting times (England)	symptoms -0.93 s.d.; employment $+3$ pp if unemployed at start	≈ -5 pp	[Oparina et al., 2024]
FE + IV: a friend's death (Australia)	1 s.d. worse mental health: -30 pp; OLS -7	-45; OLS -10	[Frijters et al., 2014]
Individual FE (UKHLS)	1 s.d. worse GHQ: -1.0 pp	-1.5 pp	[Bryan et al., 2022]
Ours: recurrence event study	symptoms $\approx +1.5$ s.d.; employment -2.7 pp	-2.7 pp	

Scaled: employment effect ÷ symptom effect (s.d.) × 1.5, the rise in symptoms at a recurrence. For the therapy studies this attributes the whole work effect to symptoms (47–77% in the Norway trial). Norway symptom effect: Knapstad et al. (2020).

▸ Fixed effects ◂ Back

Recurrence: evidence for quasi-random timing

- Clinical evidence: first episodes are provoked, later episodes increasingly autonomous (Post, 1992; Kendler et al., 1999); given clinical history and liability, the timing of a recurrence is largely unpredictable.

Effect on probability of onset (pp)	First onset	Recurrence
Death/illness of friend or relative	1.5^{**} (0.8)	3.3 (2.1)
Own illness/accident	4.8^{***} (1.4)	5.5 (4.8)
Recent job loss	0.8^{*} (0.5)	0.5 (1.1)
Separation or divorce	13.0^{***} (2.7)	5.5 (6.8)
Family problems	3.8 (4.4)	8.3 (13.3)
Problems at work	8.5^{***} (2.5)	5.9 (6.4)
Victim of crime	6.4^{*} (3.7)	-15.6 (10.7)
Controls: clinical history at $t-1$; individual FE; wave FE		

Design Back

Both directions survive individual fixed effects

Effect on the annual transition (pp)	CRE logit (paper)	Person FE	Person FE, bias-corrected
Employed last year → mental health			
Onset (healthy → depressed)	-2.2^{***} (0.7)	-3.4^{***} (0.9)	-3.8^{***} (0.9)
Recovery (depressed → healthy)	$+4.0^{***}$ (1.2)	$+3.6^{**}$ (1.4)	$+6.8^{***}$ (1.4)
Depressed last year → employment			
Exit (employed → non-employed)	$+1.1^{***}$ (0.2)	$+1.7^{***}$ (0.2)	$+1.9^{***}$ (0.2)
Entry (non-employed → employed)	-3.8^{***} (0.5)	-1.9^{*} (1.1)	-2.5^{**} (1.1)

Net of any fixed difference between men, last year's employment predicts this year's depression and vice versa. Time-varying shocks could still move both, so these remain associations.

Men 25–55, $\geq$5 waves, pre-COVID: 81,576 person-waves, 9,611 men. Controls: both lagged states, lagged earnings, children, age quadratic, wave FE. CRE: logit with person means, average marginal effects. FE: linear probability models with person FE, lagged states interacted. Bias-corrected: split-panel jackknife for Nickell bias (Dhaene & Jochmans, 2015), FE s.e. S.e. clustered by man.

◂ Job loss ◂ Recurrence

Mechanisms: why the arrows run both ways

Non-employment → depression

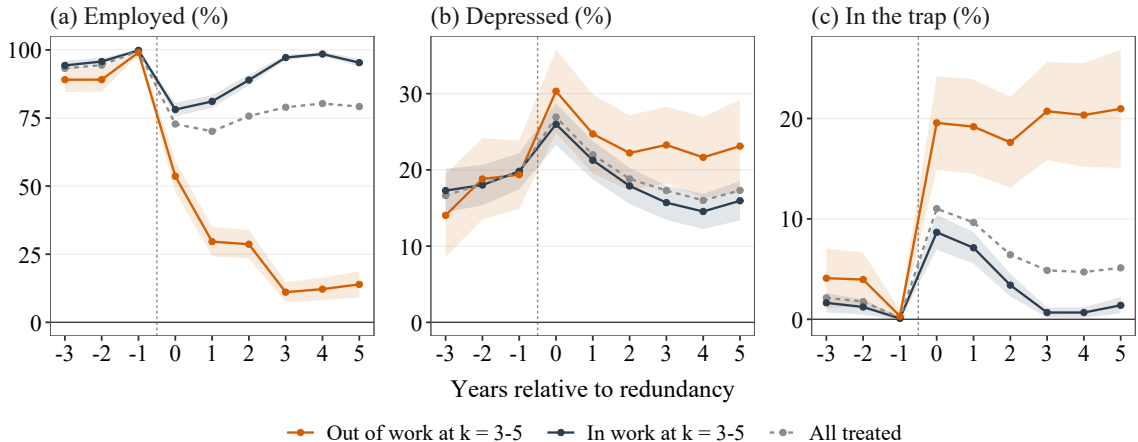
- **Behavioural activation:** less rewarding activity (Cuijpers et al., 2007)
- **Social support:** fewer daily contacts to buffer stress (Cohen and Wills, 1985)
- **Sense of purpose:** loss of time structure and status (Jahoda, 1982)

Depression → non-employment

- **Lower productivity:** job performance deficits (Adler et al., 2006)
- **Rumination:** impaired problem solving and action (Nolen-Hoeksema et al., 2008)

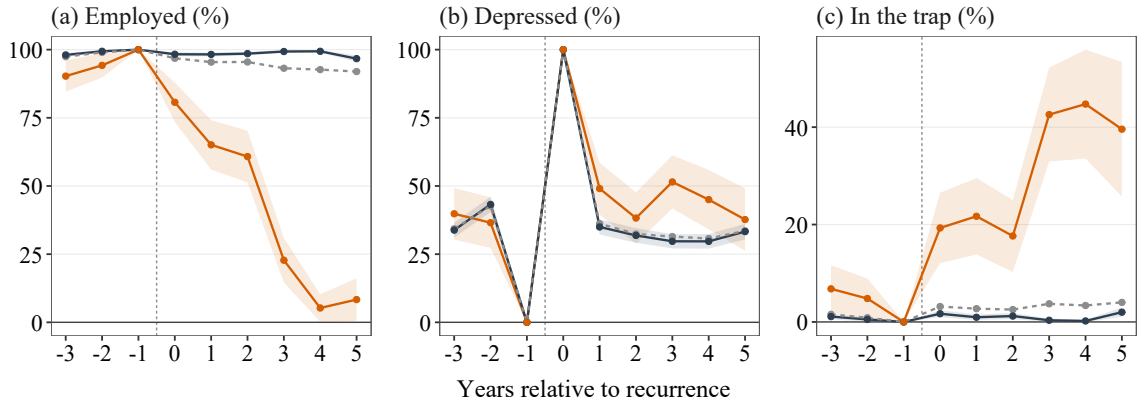
◂ Recurrence ◂ Redundancy

Redundancy: who stays out of work?



Back

Recurrence: who stays out of work?



Back

Redundancy: robustness checks

Estimators	Callaway–Sant'Anna; two-way fixed effects	▸ Estimators
Control groups	Never-redundant men with individual fixed effects; stacked clean controls	▸ Controls
Attrition	Missing years of men who leave the panel imputed (MICE)	▸ Attrition
Placebo events	Fake event dates for never-redundant men	▸ Placebos
Parallel trends	Rambachan and Roth (2023) sensitivity to violations of parallel trends	▸ Honest DiD
Genetic selection	Does the depression PGI predict redundancy?	▸ Identification

◂ Back

Recurrence: robustness checks

Clinical history	Controls with the same first episode and time in remission, who relapse later	History
Fixed effects	Individual fixed effects, pooled and stacked	Fixed effects
Control groups	All men at risk; matching also on age or on employment history	Controls
Severity	Recurrence defined at GHQ-12 caseness 3+ to 6+	Severity
Recovery and job loss	Longer remission; men seen every pre-event wave; recurrences near a redundancy dropped	Recovery
Triggers	Do provoking events predict a recurrence?	Triggers

Back

A genotype predicts the trap as well as symptoms, and for longer

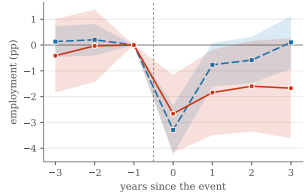
First entry into the trap, pp per s.d.	Within 1 year	Within 5 years	Within 10 years
Depression PGI	0.38^{***}	1.34^{***}	2.42^{***}
	(0.11)	(0.38)	(0.71)
GHQ-12 symptoms today	0.49^{***}	1.06^{***}	1.77^{***}
	(0.18)	(0.34)	(0.48)
Own education (fewer years)	0.14	0.38	0.54
	(0.11)	(0.37)	(0.63)
Education PGI (low)	-0.03	0.36	0.72
	(0.09)	(0.37)	(0.60)

Symptoms predict best at one year; from five years on the genotype predicts more, and its lead grows. Education and its PGI add little on their own.

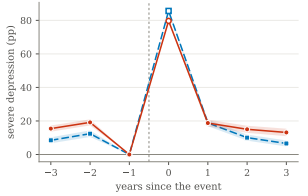
Genotyped men 25–45 in work at t, never yet in the trap; one predictor at a time; age and wave FE, ten PCs; clustered s.e.

▸ Full table ▸ Joint model ▸ All predictors ▸ Both directions ▸ No Symptoms ◂ Back

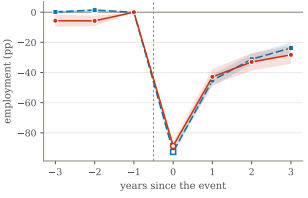
The model reproduces the event studies as untargeted moments



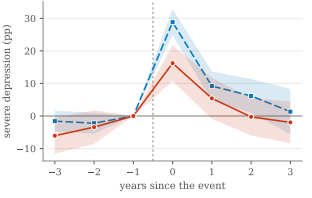
(a) Recurrence: employment



(b) Recurrence: severe depression



(c) Redundancy: employment



(d) Redundancy: severe depression

Data Model

Health fit Back

Both directions of the trap, one year ahead

State at $t+1$, pp per s.d.	Depressed	Non-employed	Both (trap)
Depression PGI	$+4.62^{***}$ (0.89)	$+0.52^{***}$ (0.18)	$+0.38^{***}$ (0.11)
GHQ-12 symptoms today	$+11.12^{***}$ (0.65)	$+0.51^{**}$ (0.21)	$+0.49^{***}$ (0.18)
Own education (fewer years)	-2.40^{***} (0.75)	$+0.50^{**}$ (0.20)	$+0.14$ (0.11)
Education PGI (low)	-1.80^{**} (0.79)	$+0.24$ (0.18)	-0.03 (0.09)
Rate (%)	14.1	1.4	0.57

Same sample and predictors as the main table, state at $t+1$: 5,964 person-waves, 940 men. The genotype and today's symptoms predict both outcomes; education predicts only non-employment.

Back

All predictors: first entry into the trap next year

pp per s.d.	Univariate	Fixed at birth	All ex-ante	All + state at t
Fixed at birth or early life				
Depression PGI	0.38^{***} (0.11)	0.37^{***} (0.11)	0.28^{**} (0.11)	0.25^{**} (0.11)
Education PGI (low)	-0.03 (0.09)	-0.13 (0.10)	-0.16 (0.11)	-0.15 (0.11)
Physical-health PGI (poor)	0.19^{**} (0.09)	0.16 (0.10)	0.17^{*} (0.10)	0.17^{*} (0.10)
Own education (fewer years)	0.14 (0.11)	0.18 (0.11)	0.08 (0.17)	0.10 (0.17)
Parental education (lower)	-0.02 (0.11)	-0.07 (0.12)	-0.08 (0.12)	-0.08 (0.12)
No professional parent (0/1)	-0.02 (0.19)	-0.08 (0.19)	-0.10 (0.21)	-0.07 (0.20)
Own history to t				
Prior GHQ-12 mean	0.29^{**} (0.13)		0.21 (0.14)	0.07 (0.17)
Ever diagnosed (0/1)	0.39 (0.52)		-0.24 (0.57)	-0.36 (0.58)
Share of working life not employed	0.18 (0.12)		0.24 (0.16)	0.24 (0.16)
Most recent earnings (low)	0.61^{*} (0.32)		0.63^{*} (0.36)	0.62^{*} (0.36)
State at t				
GHQ-12 today	0.49^{***} (0.18)			0.42^{**} (0.21)
R^2		0.010	0.018	0.020

Genotyped men 25–45 in work at t, never yet in the trap; 5,963 person-waves, 939 men; entry rate 0.57%. Age and wave FE, ten genetic PCs; s.e. clustered by man in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

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Among men with no symptoms, genetic risk still predicts the trap

State at $t+1$, pp per s.d., joint model	Depressed	Non-employed	Both (trap)
All employed men, never trapped (N = 5,964; 940 men)			
GHQ-12 at t	$+10.5^{***}$ (0.6)	$+0.49^{**}$ (0.22)	$+0.46^{**}$ (0.18)
Depression PGI	$+3.2^{***}$ (0.6)	$+0.43^{**}$ (0.18)	$+0.31^{***}$ (0.11)
Own education	-1.4^{**} (0.6)	$+0.51^{**}$ (0.20)	$+0.20^{*}$ (0.11)
Rate (%)	14.1	1.4	0.57
Employed men with GHQ-12 $= 0$ at t (N = 3,650; 821 men)			
Depression PGI	$+1.57^{***}$ (0.55)	$+0.49^{**}$ (0.19)	$+0.20^{**}$ (0.10)
Own education	-0.79 (0.54)	$+0.57^{**}$ (0.24)	$+0.18^{*}$ (0.09)
EA PGI (low)	-1.57^{***} (0.57)	-0.09 (0.22)	-0.15 (0.12)
Rate (%)	7.0	1.0	0.30

Genotyped men 25–45 in paid employment at t, never yet in the trap. Age and wave FE, number of earlier GHQ observations, 10 PCs; s.e. clustered by man.

Back

First entry into the trap: effects relative to the entry rate

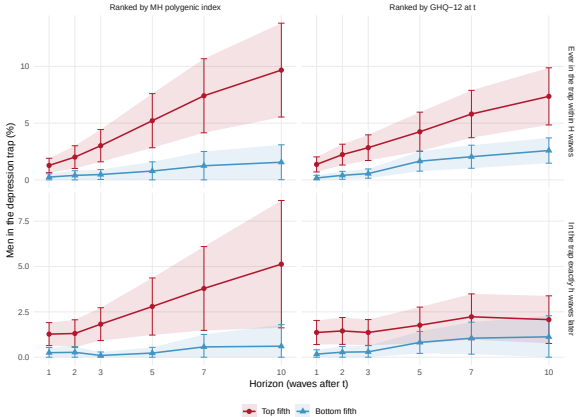
	Within 1 year		Within 5 years		Within 10 years	
First entry into the trap, per s.d.	pp	% of rate	pp	% of rate	pp	% of rate
Depression PGI	0.38***	66	1.34***	59	2.42***	60
	(0.11)		(0.38)		(0.71)	
GHQ-12 symptoms today	0.49***	87	1.06***	47	1.77***	44
	(0.18)		(0.34)		(0.48)	
Own education (fewer years)	0.14	25	0.38	17	0.54	14
	(0.11)		(0.37)		(0.63)	
Education PGI (low)	−0.03	−5	0.36	16	0.72	18
	(0.09)		(0.37)		(0.60)	
Entry rate (%)	0.57		2.26		4.00	

Relative to the entry rate, symptoms fall from 87% to 44% per s.d. as the horizon lengthens; the genotype holds at about 60%.

Genotyped men 25–45 in work at t, never yet in the trap; 964 men, 5,964–6,425 person-waves; one predictor at a time; age and wave FE, ten PCs; s.e. clustered by man.

Back

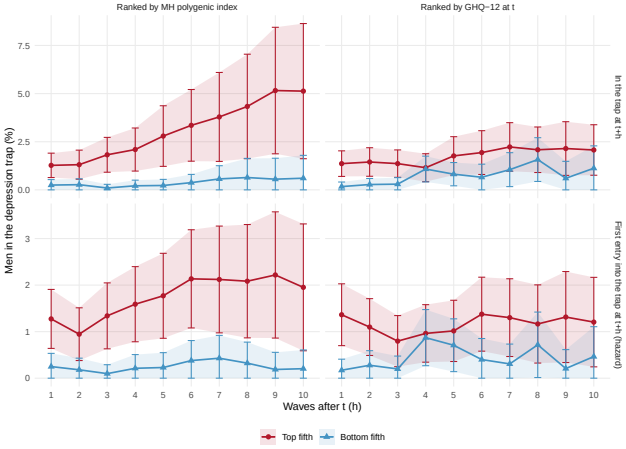
Over a decade the genetic ranking keeps opening; the symptom ranking stops



Genotyped men aged 25–45 in paid employment at t, never observed in the trap before. Top and bottom fifth of each predictor (ties broken at random; GHQ top fifth = score of 2 or more, bottom fifth = 0). 95% intervals, standard errors clustered by man.

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Year by year: the genetic hazard holds, the symptom hazard fades



Genotyped men aged 25–45 in paid employment at t, never observed in the trap before. Top and bottom fifth of each predictor (ties broken at random; GHQ top fifth = score of 2 or more, bottom fifth = 0). Hazard: entry at t+h among men not in the trap at any observed wave between t and t+h−1. 95% intervals, standard errors clustered by man.

The same four predictors, entered jointly

	Within 1 year		Within 5 years		Within 10 years	
First entry into the trap, per s.d.	pp	% of rate	pp	% of rate	pp	% of rate
Depression PGI	0.31***	55	1.17***	52	2.13***	53
	(0.11)		(0.38)		(0.70)	
GHQ-12 symptoms today	0.46**	81	0.93***	41	1.51***	38
	(0.18)		(0.34)		(0.49)	
Own education (fewer years)	0.20*	35	0.36	16	0.47	12
	(0.11)		(0.38)		(0.65)	
Education PGI (low)	−0.09	−16	0.20	9	0.46	11
	(0.10)		(0.37)		(0.61)	
Entry rate (%)	0.57		2.26		4.00	
Person-waves	5,964		6,409		6,425	

Genotyped men 25–45 in work at t, never yet in the trap; 964 men. Age and wave FE, number of earlier GHQ observations, ten PCs; s.e. clustered by man.

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Position in the literature: full citations

1. Economics of mental health

Mental health and work: Ettner et al., 1997; Frank and McGuire, 2000; Frijters et al., 2014; Bryan et al., 2022; Biasi et al., 2021; Ridley et al., 2020; Paul and Moser, 2009; Kuhn et al., 2009

Job loss, work and therapy: Marcus, 2013; Picchio and Ubaldi, 2024; Sterud et al., 2025; Hussam et al., 2022; Smith et al., 2025

⇒ persistence of the depression trap (depressed and non-employed) after recurrences and redundancies

2. Genoeconomics

Polygenic indices and outcomes: Biroli et al., 2022; Benjamin et al., 2024; Okbay et al., 2022; Papageorge and Thom, 2020; Barth et al., 2020; Amin et al., 2019; Martin Bassols et al., 2024

Genetics in structural models: Barth et al., 2022; Houmark et al., 2024; Bolt et al., 2025; Agostinelli and Weingarten, 2025; Yang et al., 2026

⇒ genetic risk for depression predicts employment dynamics and the employment response to shocks

▸ Strands 3–4 ◂ Back

Position in the literature: full citations (continued)

3. Structural models of health and labor supply

Health, disability and work: French, 2005; French and Song, 2014; Low and Pistaferri, 2015; De Nardi et al., 2016; Hosseini et al., 2021; De Nardi et al., 2024; Capatina and Keane, 2026

Mental health: propagation of shocks [Jolivet and Postel-Vinay, 2025; Diaz-Campo et al., 2025; Drozd, 2026]; endogenous therapy [Cronin et al., 2024; Abramson et al., 2024; Sauer et al., 2025]

$\Rightarrow$ the first structural evaluation of the fiscal case for targeting and expanding mental health care; genetic risk as an ex-ante predictor; employment decisions reinforce mental-health shocks

4. Public economics of health care

Tagging and prediction: Akerlof, 1978; Mankiw and Weinzierl, 2010; Logue and Slemrod, 2008; Kleinberg et al., 2015; Kitagawa and Tetenov, 2018; Azevedo et al., 2026

Rationing by waiting: Lindsay and Feigenbaum, 1984; Gravelle and Siciliani, 2009; Aakvik et al., 2015; Godøy et al., 2024; Prudon, 2026; Dodd et al., 2025

Fiscal returns to (mental) health care: Hendren and Sprung-Keyser, 2020; Layard et al., 2007; Oparina et al., 2024; Serena, 2025

$\Rightarrow$ what a genetic tag adds to allocating rationed therapy, beyond observed histories, with UK taxes and benefits

Back

Data: measurement and sample selection

Dataset: UK Household Longitudinal Study + BHPS (1991–2023)

- Panel linking labor market outcomes, health, and **molecular genetics**.
- **MH measure:** GHQ-12 caseness, with 2+ symptoms = mild, 4+ = probable case (Goldberg et al. 1997; Hardy et al. 1999). ▸ GHQ-12
- **Genetic risk:** depression PGI (Baselmans et al., 2019), oriented so *higher = higher risk*; external GWAS weights, no sample overlap.

Sample Selection

- **Focus:** Working-age men (25–55) with valid genetics data and labor market status*.
- **Model Sample:** Extends to (25–65) and includes retirees (65–80).
- **Full sample:** ≈29,000 men, ≈160,000 person-waves (no genetics)
- **Genotyped:** ≈2,200 men, ≈21,000 person-waves (of ≈10,000 genotyped)
- **Women:** qualitatively similar results; deferred to follow-up work to avoid modelling fertility jointly with health and labor supply.

**Status includes: Employed, Unemployed, or Long-term sick/disabled.*

▸ Women ◂ Back

GHQ-12 Survey: “Have you recently…”

- been feeling unhappy or depressed? (depressed mood)
- been feeling reasonably happy, all things considered? (mood)
- been able to enjoy your normal day-to-day activities? (anhedonia)
- lost much sleep over worry? (insomnia)
- been thinking of yourself as a worthless person? (worthlessness)
- been able to concentrate on whatever you’re doing? (concentration)
- felt capable of making decisions about things? (indecisiveness)
- felt constantly under strain?
- felt you couldn’t overcome your difficulties?
- been able to face up to problems?
- been losing confidence in yourself?
- felt that you were playing a useful part in things?

Depressive symptoms

Distress, coping and functioning

GHQ-12 caseness is supported by clinical validation.

At the 4+ cutoff, GHQ-12 identifies **69% of clinical-interview cases** and correctly classifies **88% of non-cases** (Hardy et al., 1999).

Further validation: Goldberg et al. (1997); Donath (2001); Lundin et al. (2016); Anjara et al. (2020).

▸ Validity ◂ Back

How well does GHQ-12 caseness track a clinical diagnosis?

Study	Setting, *N*	Clinical reference	AUC	Sensitivity / specificity
Goldberg et al. (1997)	WHO primary care, 15 centres; 5,438	CIDI-PC diagnosis	0.88 avg. (0.83–0.95)	GHQ-12 as good as GHQ-28
Hardy et al. (1999)	**England**, NHS employees; 551	CIS-R psychiatric interview	—	**0.69 / 0.88 at 4+** (best threshold)
Donath (2001)	Australia, population; 10,641	CIDI diagnosis	0.78	0.75 / 0.70 at 1+
Lundin et al. (2016)	Sweden, population; 484	Major or minor depression (DSM-III-R)	0.73	0.67 / 0.74 at 2+
Anjara et al. (2020)	Indonesia, primary care; 676	CIS-R, any ICD-10 disorder	0.79 (mood: 0.70)	0.82 / 0.64 at 2+

- Screens for **common mental disorder** (depression and anxiety); accuracy for depression alone is lower (AUC 0.70–0.73)
- UK convention: **4+ = probable case**, the best threshold in the UK validation (Hardy et al. 1999); optimal thresholds rise with prevalence (Goldberg et al. 1998)
- Answers are relative to “usual”, so standard scoring can miss chronic symptoms (Goodchild & Duncan-Jones 1985)

Back

Implementation considerations

- **Administrative burden.** Under fixed capacity nothing new is bought: an existing queue is re-ordered. The elastic arm prices the capacity expansion separately (£488–841 per course).

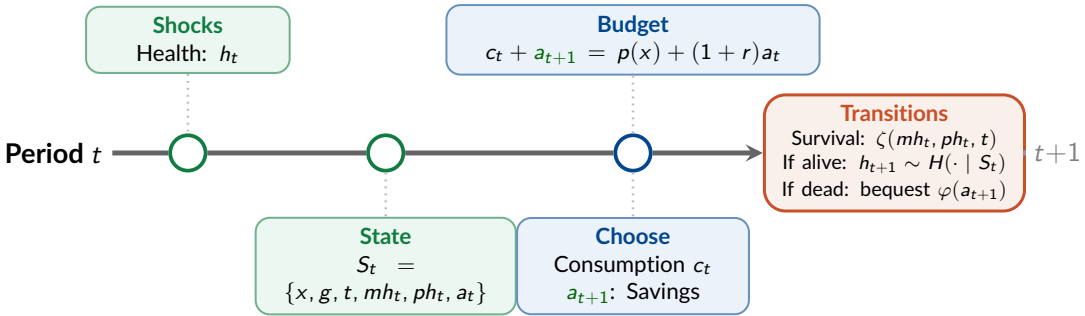
- **Clinical framing.** Risk should be presented as non-deterministic (Lebowitz and Ahn, 2018) and embedded in preventative care rather than attached to a psychiatric label (Austin, 2020).

- **Equity.** PGI portability is limited across ancestry groups, and under fixed capacity priority for one group means delay for another.

- **Signal quality.** The MDD polygenic index is projected to approach its ˜30–50% R^2 heritability ceiling as GWAS samples grow, so the return to targeting should rise over time.

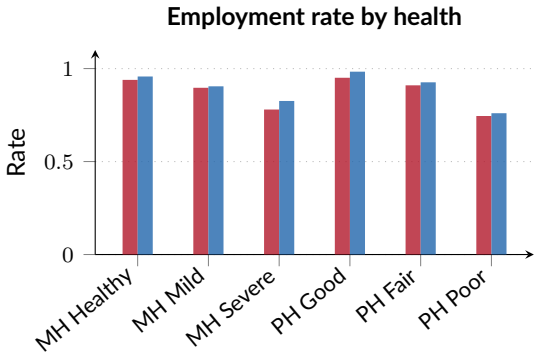
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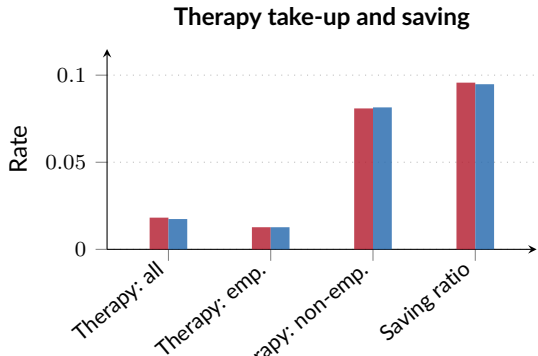
Retirement timeline



Back

Model fit: employment, therapy and savings

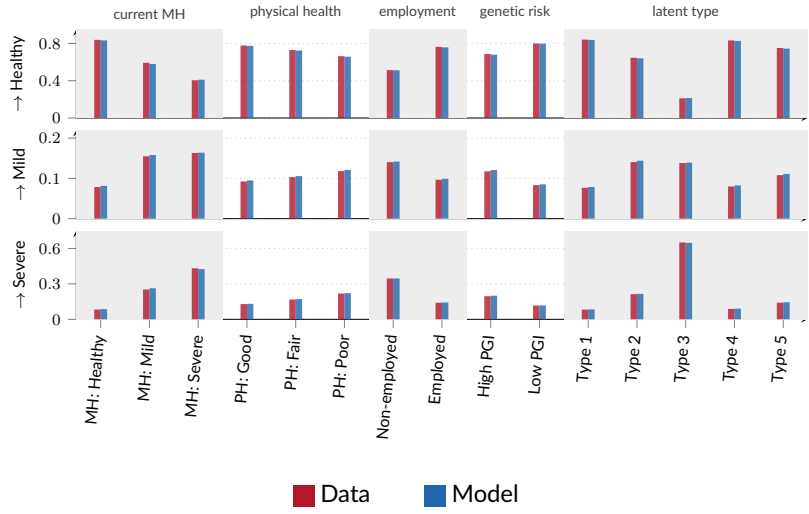




Data Model

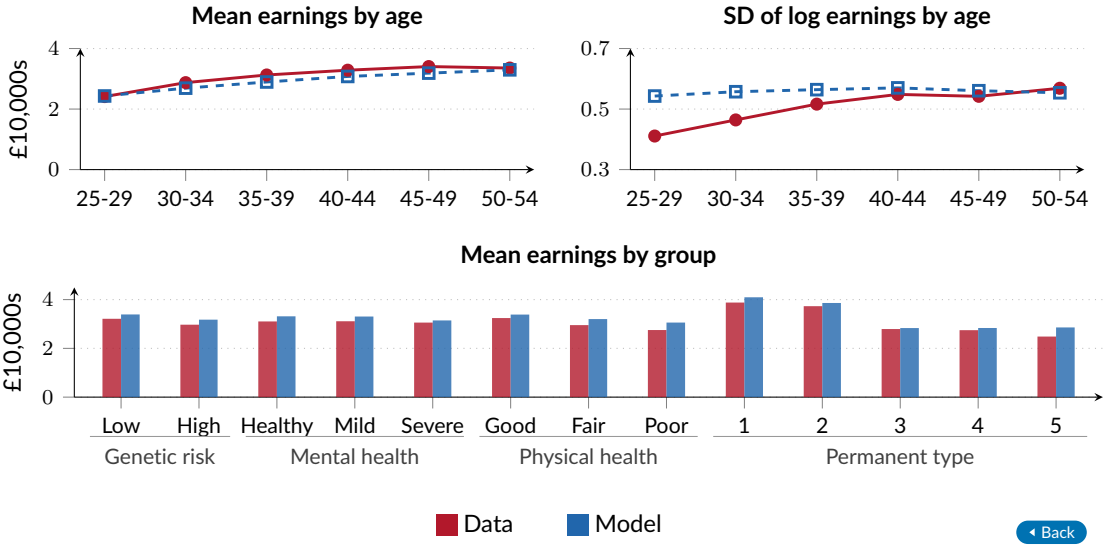
Back

Model fit: mental-health transitions



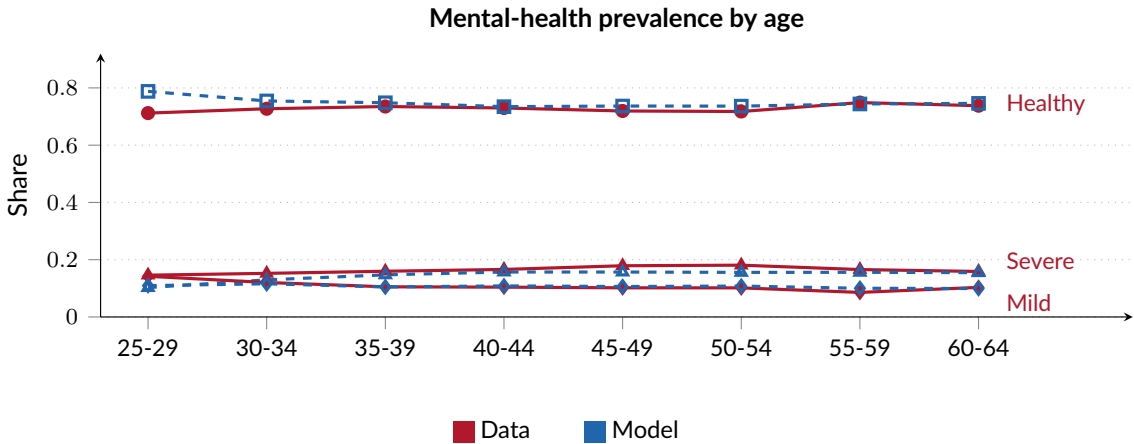
Back

Model fit: earnings



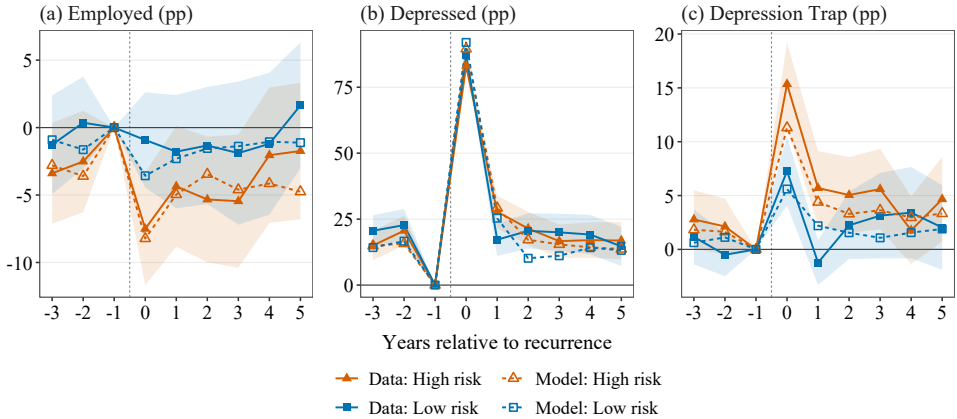
Back

Model fit, untargeted: prevalence by age



Back

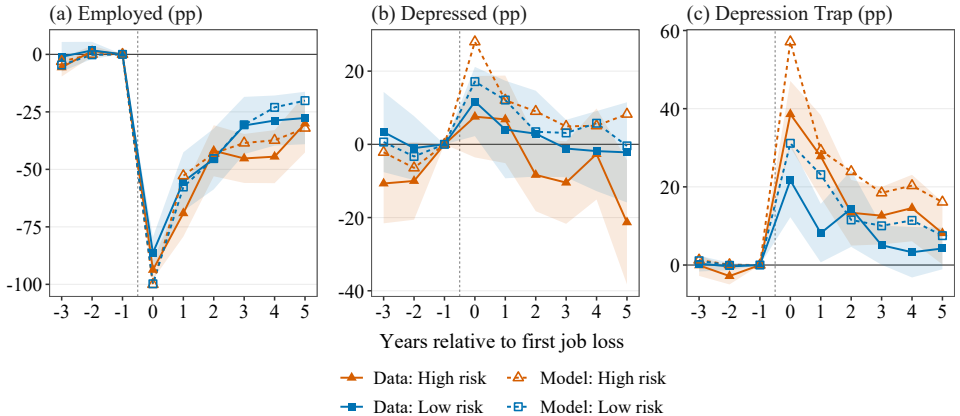
By genetic risk: the model reproduces the recurrence gradient



At impact, employment: -8.2 vs. -3.6 pp in the model, -7.5 vs. -0.9 in the data

Job loss by risk Back

By genetic risk: job loss

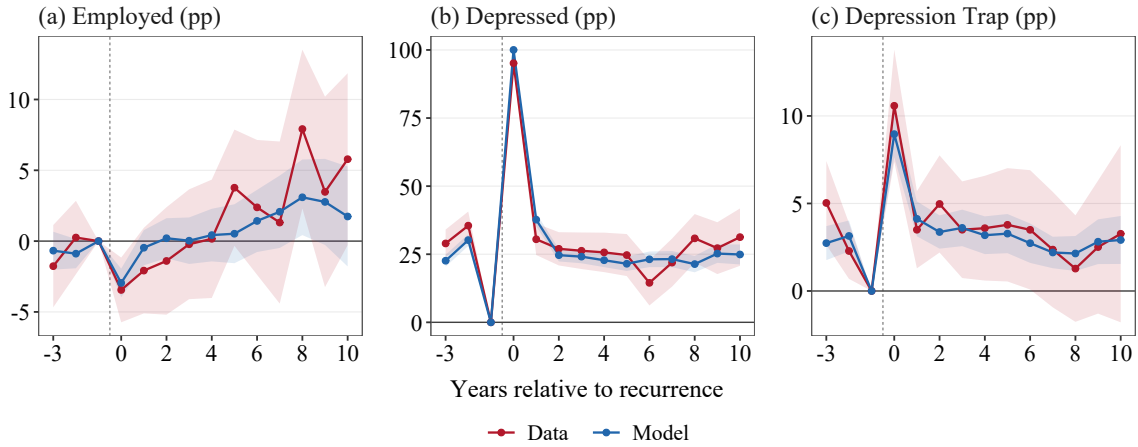


High- minus low-risk trap gap, $k = 1$–5: $+8.3$ pp in the data, $+8.9$ in the model

Depression does not match: in the data, high-risk men are already depressed before the loss.

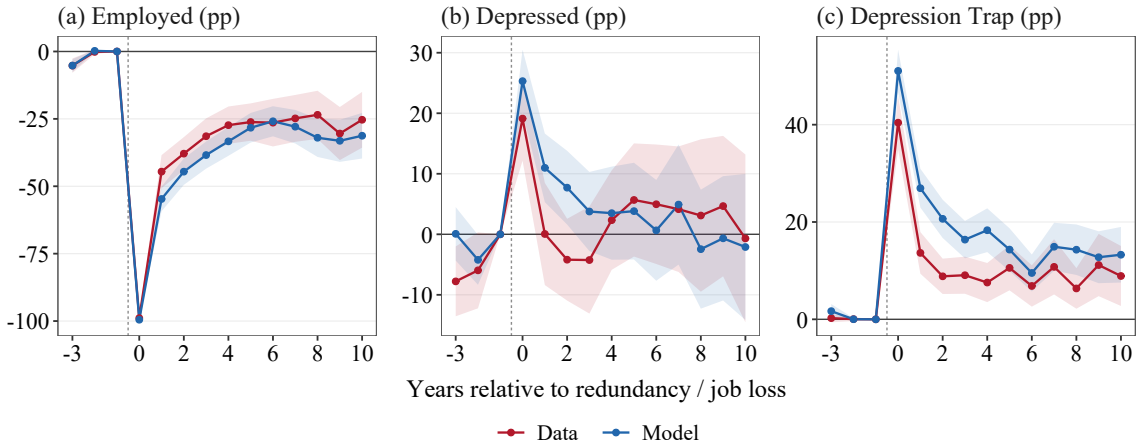
▸ Recurrence by risk ◂ Back

Untargeted Model Fit: recurrence event study



Back

Untargeted Model Fit: redundancy event study



Back

The classification recovers the model's own types

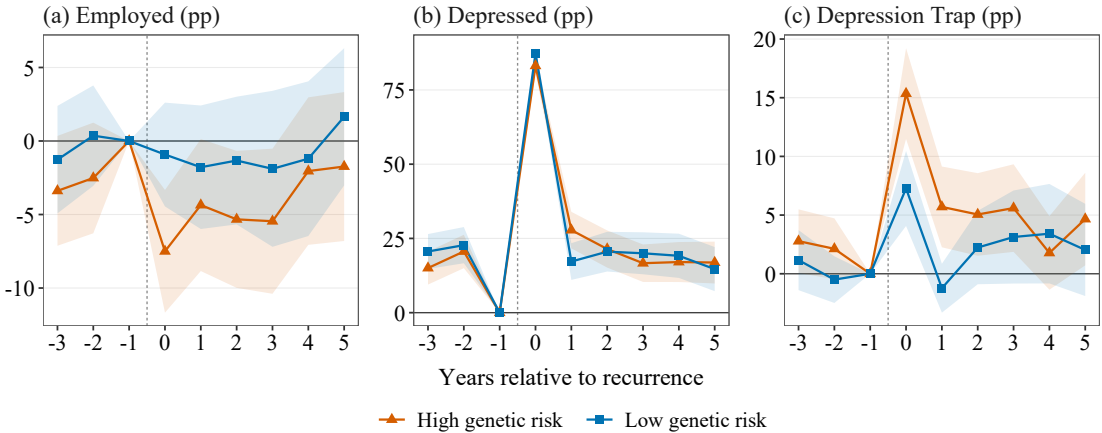
$\rightarrow$ The same k-means, run on the model's simulated panel with the same person-level moments as in the data: health histories and the wage predicted at age 40.

	Recovered type (% of row)				
True type	1	2	3	4	5
Type 1	**95.5**	4.5	0.0	0.0	0.0
Type 2	14.4	**85.3**	0.1	0.1	0.2
Type 3	0.0	0.0	**98.9**	0.6	0.5
Type 4	0.0	0.5	0.1	**98.6**	0.9
Type 5	0.0	0.1	0.2	1.4	**98.3**

95.4% of simulated men are assigned to their true type. The only real confusion is Type 2 read as Type 1, the two types with the smallest permanent wage gap.

Back

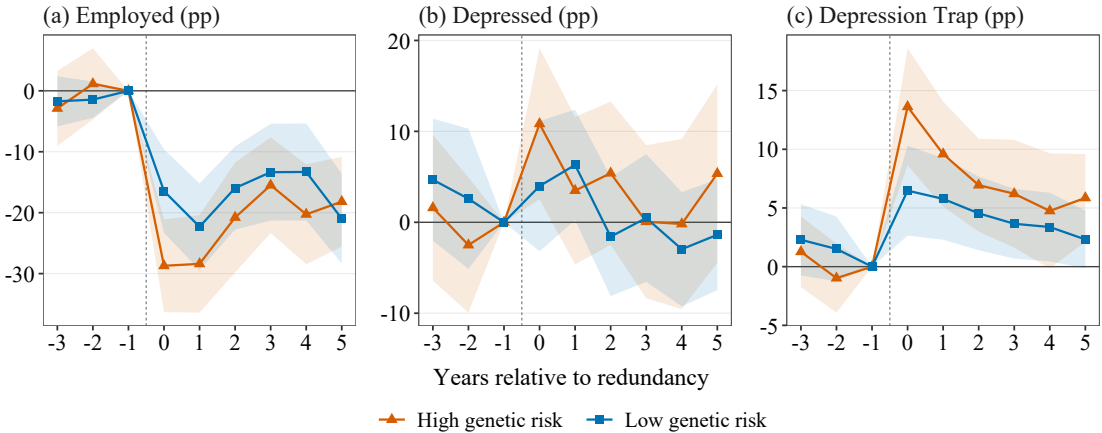
Recurrence by genetic risk



Employment at impact: -7.5 vs. -0.9 pp

Redundancy G×E
Back

Redundancy by genetic risk



Employment at impact: -29 vs. -17 pp

▸ Recurrence G×E ◂ Back

The depression trap: evidence from the literature

- **Employment → Depression**
  - Longitudinal meta-analysis, controlling for income: 0.19 SMD (Sterud et al., 2025)
  - Exogenous variation (e.g. plant closures): ∼0.15 SMD (Picchio and Ubaldi, 2024)
  - RCT, work vs. cash of equal value: 0.23 SD non-pecuniary effect on depression (Hussam et al., 2022)
  - *Reference*: antidepressants 0.3 SMD, psychotherapy 0.6 SMD
  - *Mechanisms*: social support, status, purpose, optimism (Fossey and Harvey, 2010)
- **Depression → Employment**
  - One s.d. worse MH ⇒ 7–30 pp drop in employment (Ettner et al., 1997; Frijters et al., 2014; Smith et al., 2025)
  - *Mechanisms:* cognitive impairment, reduced motivation, pessimism about 'returns to effort' (Ridley et al., 2020; De Quidt and Haushofer, 2016)
- **Both directions present in our panel** (CRE logit on the full panel, net of permanent heterogeneity & past income): employed ⇒ -2.2 pp depression onset, $+4.0$ pp recovery; depressed ⇒ $+1.1$ pp employment exit, -3.8 pp re-entry.
- Two-way feedback admits a self-reinforcing state: **out of work and depressed**. Does genetic risk govern who enters it — and who stays?

▸ Econometric Specification ◂ Back

Identification: two shocks, one for each direction

1. Redundancy

Employment → Mental Health

- Firm-side job loss: the *position* is eliminated, worker not at fault (plant-closure tradition, Picchio and Ubaldi, 2024)
- No polygenic index predicts redundancy ($p \geq 0.22$) — but all three predict *health-related* job endings ($p < 0.001$)
- Benchmark effect sizes: job loss ∼0.10 s.d. of MH (meta); RCT value of work 0.23 s.d. (Hussam et al., 2022)

2. Recurrence

Mental Health → Employment

- First onset is *provoked* — often by the very job loss we study (Kendler et al., 1999)
- *Kindling*: recurrences need little provocation; timing near-unpredictable given clinical history (Post, 1992)
- To our knowledge, first use of recurrence timing as identifying variation

Two independent sources of variation ⇒ largely non-overlapping assumptions.

Back

Trap dynamics: econometric specification

First-order dynamic logit per transition, estimated *separately by genetic risk group* (median PGI split), so every coefficient is free to differ across groups:

$$\Pr(Y_{it} = 1 \mid G_i = g) = \Lambda\big(\rho_g\, Y_{i,t-1} + \mathbf{X}'_{i,t-1}\gamma_g\big)$$

- **Controls $\mathbf{X}_{i,t-1}$:** lagged other state (depression / employment), lagged earnings, children, age quadratic, wave FE, ten genetic PCs
- **Estimates:** average predicted transition probabilities, both groups' fits averaged over the *same* pooled at-risk covariate distribution $\Rightarrow$ differences are behaviour, not composition
- **Inference:** SEs clustered by individual; SE(diff) $= \sqrt{se^2_{\text{low}} + se^2_{\text{high}}}$ (disjoint groups)
- **Sample:** genotyped men 25–55, $\geq$5 waves, pre-COVID: 15,206 person-waves, 1,534 men; depression = GHQ-12 caseness ≥ 4
- **Why no person-means:** a Mundlak/CRE variant nulls every gradient — genetic risk is time-invariant, and person-means of its consequences absorb the identifying between-person variation (paper Table A2). Contemporaneous feedback is left to the structural model.

Types Back

Robustness: the gradient is not a gradient in permanent type

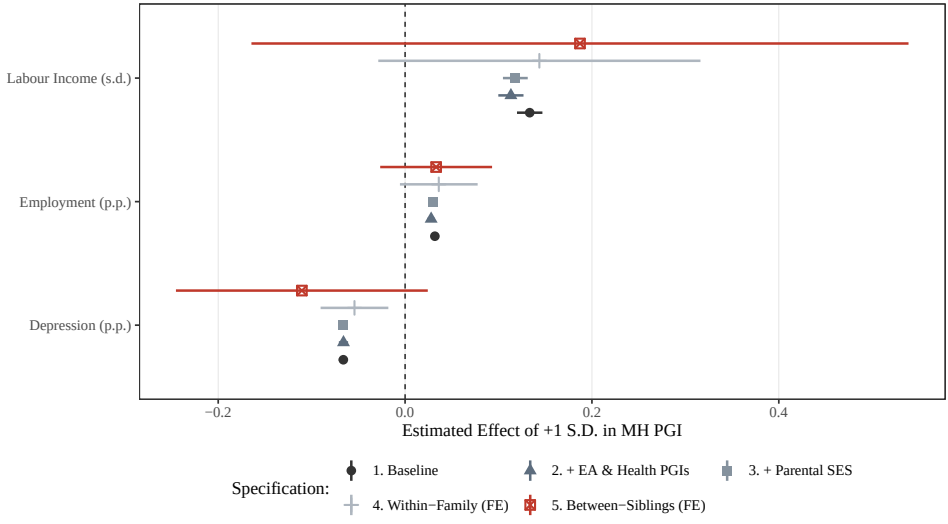
Difference in annual transition probability, high − low risk (pp)	Split on type-clustering PGI	
	Baseline	+ type dummies
Mental health		
Onset (healthy → depressed)	$+6.3^{***}$	$+6.6^{***}$
Recovery (depressed → healthy)	-14.5^{***}	-13.4^{***}
Employment		
Exit (employed → non-employed)	$+0.8^{***}$	$+1.0^{***}$
Entry (non-employed → employed)	-9.1^{*}	-9.2^{**}
"Depression trap"		
Entry into trap	$+1.0^{***}$	$+1.0^{***}$
Exit from trap	-21.3^{***}	-17.9^{**}
Expected trap spell, low / high risk (yrs)	1.7 / 2.7	1.8 / 2.7

Adding the model's five types leaves every margin intact: only trap exit moves. Types are residualised on this index, so they are orthogonal to the split by construction.

Same 1,534 men; types are the model's x_i. Paper Table A8.

Back

Within-family robustness: PGI stability



Back

Women: the penalty replicates on a different margin

	Men		Women	
Effect of $+1$ s.d. of genetic risk	Est.	% via MH	Est.	% via MH
Probable major depression (pp)	6.59^{***}	—	8.31^{***}	—
Employed (pp)	-2.61^{***}	40	-4.01^{***}	37
Long-term sick/disabled (pp)	1.61^{***}	53	2.48^{***}	45
Looking after home/family (pp)	0.37^{**}	6	1.19^{***}	32
Labor income (s.d.)	-0.112^{***}	24	-0.078^{***}	26
Log gross pay, if employed	-0.054^{***}	7	-0.054^{***}	4

On the extensive margin the penalty is larger for women, and mediation through mental health is comparable. This is not a power difference: 3,540 genotyped women against 2,605 men.

What differs is where the withdrawal goes: family care absorbs $\sim$30% of women's employment penalty, a destination men essentially do not enter.

Broad working-age sample (all activity statuses), 25–55, pre-2020; men re-estimated on the same sample, so the men's column differs slightly from the main text.

Dynamics Back

Women: the transition gradients replicate

Annual transition probability (%)	Women			Men
	Low risk	High risk	Diff.	Diff.
Mental health				
Onset (healthy → depressed)	14.0	18.7	$+4.7^{***}$	$+3.5^{***}$
Recovery (depressed → healthy)	54.9	46.1	-8.7^{***}	-15.3^{***}
Employment				
Exit (employed → non-employed)	3.5	4.3	$+0.9^{***}$	$+0.7^{**}$
Entry (non-employed → employed)	21.1	17.3	-3.8^{**}	-13.7^{***}
"Depression trap"				
Entry into trap	2.7	4.3	$+1.7^{***}$	$+0.7^{***}$
Exit from trap	49.9	42.2	-7.7^{**}	-24.4^{***}

All six gradients hold, and trap entry is more strongly graded for women than for men.

Trap persistence, the margin the model is built around, is a third the size and fragile: dropping family care from the state set reduces it to -1.7 (s.e. 5.6). For women, "non-employed and depressed" is partly a fertility choice.

Fertility Back

Women: genetic risk predicts completed fertility

Effect of $+1$ s.d. of genetic risk	Mean	Women	Men
First-birth hazard, childless 20–45 (pp)	6.8	-0.18	-0.23
Children in household, ages 40–45	1.32	-0.094^{***}	-0.020
Childless at ages 40–45 (pp)	26.7	$+3.20^{***}$	$+1.52$

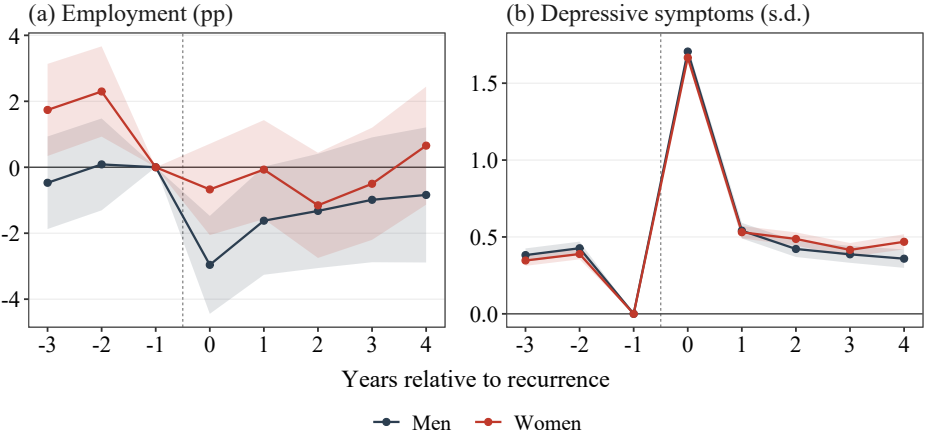
Genetic risk moves completed fertility while leaving the timing of a first birth untouched: the hazard is a precise null, but there are 0.09 fewer children and 3.2 pp more childlessness per s.d. There is no comparable effect for men.

Nor can it be controlled away. Conditioning on children amplifies rather than attenuates the women's penalty: employment $-4.01 \rightarrow -4.21$, family care $1.19 \rightarrow 1.39$. A confounder shrinks on conditioning; a consequence need not.

$\Rightarrow$ Fertility has to be modelled as a choice responding to the same genetic risk, which is a strictly larger model than this paper's.

▸ Redundancy ◂ Back

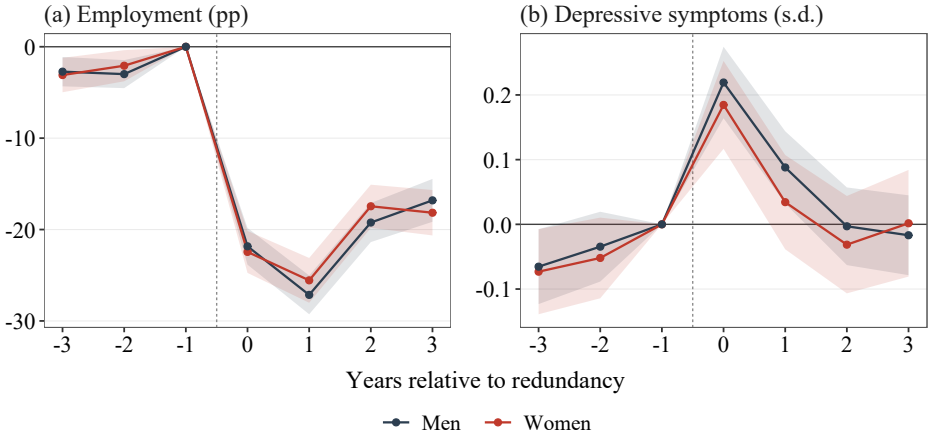
Women: the recurrence design does not transfer



$\rightarrow$ The same clinical event: symptoms $+1.7$ s.d. in both samples.
Employment falls only for men, -3.0 pp against a precise null for women.

Redundancy Back

Women: the redundancy design transfers



→ Employment -22.4 pp (women) vs -21.8 (men); symptoms $+0.18$ vs $+0.22$ s.d. Pre-trends are comparable in both samples.

Recurrence Back

Estimation strategy

1. **Step 1: Classification (Unobserved Heterogeneity)**
   - **Method:** K-means clustering on wages, mental and physical health, residualised on genetics, following Bonhomme et al. (2022).
   - **Output:** Assigns every individual a discrete type x.
2. **Step 2: External Estimation (Direct from Data)**
   - **Physical health** transitions (π): depend only on predetermined covariates.
   - Survival (life tables), Pensions ($p(x)$), Benefits ($b(n, mh, ph)$), Tax schedule $\mathcal{T}(\cdot)$, Therapy Cost (P_τ), Receipt Probability (λ_τ, calibrated using administrative data).
   - **Calibrated:** $\omega = 3$, $r = 4.6\%$, $\ell_n = 0.4$, bequests (θ, κ; Maccuish 2022).
3. **Step 3: Internal Estimation (SMM)**
   - **Mental health** transitions ($\mu_1, \mu_3 - \mu_5$): therapy is endogenous, so estimated inside the model; therapy effect μ_2 calibrated from RCTs.
   - **Wage Process:** gradients by health/type/gene, human capital return, shocks (ρ, σ_η).
   - **Preferences:** Disutility of Labor ($\delta(mh, ph)$) and Therapy ($\psi(n)$), Discount Factor (β), Consumption Share (γ).
   - **Target:** employment, savings, wages, treatment rates, MH transition rates.

▸ Parameters ◂ Back

Type characteristics

	Share	Mental health (%)			Physical health (%)			Employed	Average…	
Type	(%)	Healthy	Mild	Severe	Healthy	Average	Poor	(%)	Earnings (£)	Age
1	20.1	83.1	8.9	8.0	65.9	25.3	8.8	97.3	41,800	44
2	17.5	64.1	13.1	22.8	68.9	23.6	7.5	96.8	39,300	46
3	9.1	17.9	11.9	70.2	41.7	30.0	28.3	65.5	28,600	46
4	33.0	82.5	9.1	8.4	70.0	23.4	6.6	94.2	27,800	47
5	20.2	74.1	11.6	14.3	20.3	37.1	42.7	80.9	25,700	46

- **1:** good health, highest earnings **2:** high earnings, more depression
- **3:** mostly severely depressed, least employed
- **4:** good health, lower earnings **5:** poorest physical health, lowest earnings

k-means ($K = 5$) on wage and health histories (Bonhomme et al., 2022); men 25–64, not retired: 23,051 person-waves, 2,651 men. Shares are person-level; earnings are mean real gross annual earnings of the employed (2015 prices).

Clustering moments Back

Residual Permanent Heterogeneity: Type

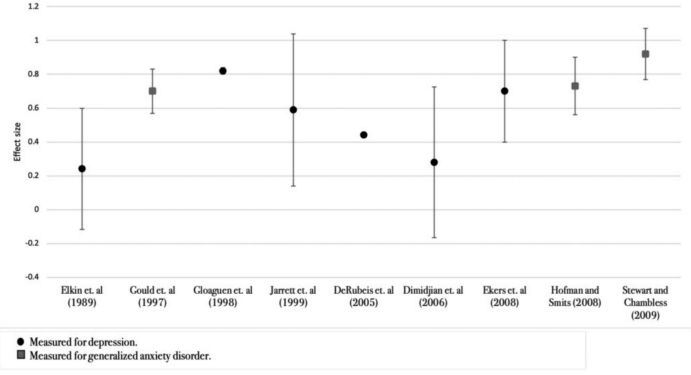
- Persistent differences in productivity and health endowments beyond the measured genetic index.
- 5 types from k-means clustering of wage and health moments (Bonhomme et al., 2022).
- Moments are residualised on the depression PGS and adjusted for age and cohort.

	Time in poor health (% of years)			% of men		PGS (s.d.)	Health behaviours (%)		
Type	Poor MH	Poor PH	Sick/disabled	Degree	Top occ. class	Self-rated health	Obese	Smoker	No sport
1	8	9	1	67	35	$+0.07$	13	7	5
2	8	7	1	2	9	$+0.04$	20	22	13
3	23	8	1	51	33	$+0.07$	18	13	8
4	70	28	25	16	18	-0.12	30	37	25
5	14	43	13	3	8	-0.13	32	36	22

MH / PH: mental-health / physical-health
Sick/disabled: long-term sick or disabled
Top occ. class: higher managerial and professional occupations

▸ Type characteristics ◂ Back

Therapy: Effects on Mental Health



Standardised effects of a course of talk therapy, with 95% confidence intervals where reported. Source: Cronin et al. (2025), Figure 2.

- **Therapy:** $\approx$ **0.6** s.d.
  6–9 months: 0.74 s.d.; 10–12 months: 0.49 s.d. Cuijpers et al. (2023)
- **Antidepressants:** $\approx$ 0.3 s.d. Cronin et al. (2025)

Back

Estimated structural parameters

Wage Process Estimates

Parameter	Est.
β_0: Intercept	-0.060
β_{age}: Age (Linear)	0.041
β_{age^2}: Age (Squared)	-0.067
β_g: Genetic Return	0.071
Health Returns	
$\beta_{MH,1}$ (Good MH)	0.154
$\beta_{MH,2}$ (Poor MH)	0.138
$\beta_{PH,1}$ (Good PH)	0.211
$\beta_{PH,2}$ (Poor PH)	0.017
Shocks	
ρ: Persistence (AR1)	0.894
σ_η: Shock Std. Dev.	0.410

Utility & Cost Estimates

Parameter	Est.
β: Discount Factor	0.951
γ: Consumption Share	0.527
ω: Risk Aversion	3.000
Disutility of Labor (δ)	
Mental Health (Good)	0.023
Mental Health (Poor)	0.047
Mental Health (Severe)	0.078
Physical Health (Good)	0.031
Physical Health (Poor)	0.059
Physical Health (Severe)	0.070
Treatment Costs (ψ)	
Employed	0.029
Non-Employed	0.019

Back

The model in one slide: work and health feed back on each other

From 25 to 64, a man in state $S_t = (t, a_t, hc_t, \epsilon_t, mh_t, ph_t, g, x)$ chooses each year whether to **apply for therapy**, whether to **work** and how much to **save**; at 65 he retires.

$$V_t(S_t) = \max_{d_t \in \{0,1\}} \mathbb{E}_{\nu_t}\Big[\max_{n_t,\, a_{t+1}} \Big\{ U(c_t, n_t, \tau_t; mh_t, ph_t) + \beta\, \mathbb{E}_t\big[V_{t+1}(\underbrace{a_{t+1}, hc_{t+1}, mh_{t+1}, ph_{t+1}}_{\text{endogenous}}; \underbrace{t{+}1, \epsilon_{t+1}, g, x}_{\text{exogenous}}) \big] \Big\} \Big]$$

Preferences $U = \dfrac{(c_t^{\gamma} L_t^{1-\gamma})^{1-\omega}}{1-\omega} - \psi(n_t)\,\tau_t, \qquad L_t = 1 - \ell_n n_t - \delta(mh_t, ph_t)$

Budget $c_t + a_{t+1} = (1+r)a_t + n_t w_t\,[1 - \mathcal{T}(\cdot)] + b(n_t, mh_t, ph_t) - P_\tau(n_t)\,\tau_t, \quad a_{t+1} \geq 0$

Wages $\ln w_t = \beta_1 g + \beta_2 x + \beta_3' mh_t + \beta_4' ph_t + f(hc_t) + \epsilon_t, \qquad hc_{t+1} = (1-\delta_{hc})\,hc_t + n_t$

Health $mh^*_{t+1} = \mu_1' mh_t + \mu_2\,\tau_t + \mu_3 n_t + \mu_4' ph_t + \mu_5'(g, x, t) + \upsilon_{t+1}$ $\quad ph_{t+1}$ alike, without τ_t

ordered logit

Therapy applicants are treated with probability λ_τ: $\tau_t = d_t \nu_t$, $\nu_t \sim \text{Bern}(\lambda_\tau)$

From 65 no work or therapy; pension $p(x)$, survival $\zeta(mh_t, ph_t, t)$, bequests

health → work, today work → health and skills, next year genetic risk

▸ Health ▸ Retirement ◂ Back

Health dynamics

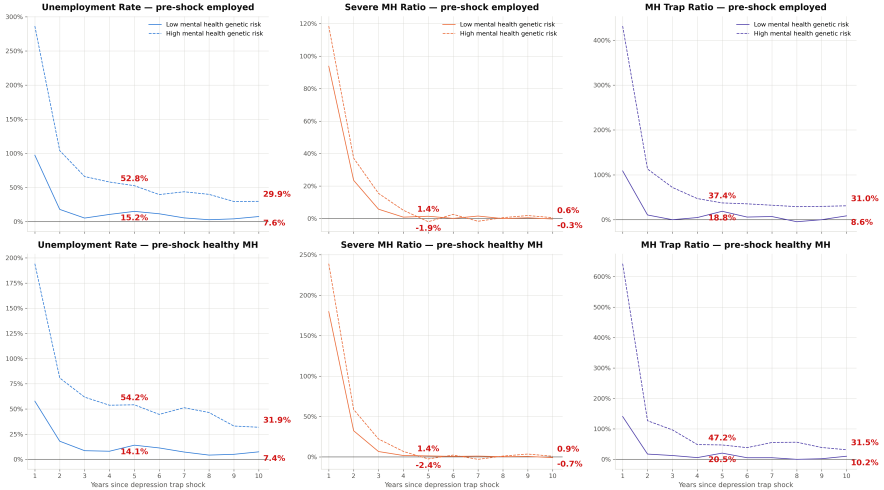
Health Transitions (Ordered Logit): Future mental health is determined by a latent index:

$$mh^*_{t+1} = mh_t\mu_1 + \underbrace{\tau_t\,\mu_2}_{\text{Therapy}} + \underbrace{n_t\,\mu_3}_{\text{Work}} + ph_t\mu_4 + \underbrace{S_t^{exo}\mu_5}_{\text{age, type } x\text{, genetics } g} + \upsilon_{t+1}$$

- **Explicit Process:** mh_{t+1} is determined by where mh^*_{t+1} falls relative to cut-points κ_j (proportional odds).
- **Feedback:** Employment (n_t) improves the latent mental health index.
- **Therapy effect** μ_2**:** calibrated from RCT meta-analyses (SMD ≈ 0.6, converted to log-odds), because treatment is endogenous.
- **Estimation asymmetry:** the physical health process depends only on predetermined covariates $\Rightarrow$ estimated *outside* the model; the mental health process depends on the endogenous therapy decision $\Rightarrow$ μ_1, μ_3–μ_5 estimated *inside* the model via SMM.
- **Symmetric PH Process:** Physical health follows the same structure, without a therapy term.

Back

Trap shock: conditioning on the pre-shock state



→ The divergence is not compositional: it holds among men employed before the shock (top) and among those in good mental health before it (bottom).

Back

Expanding the therapist workforce: 5, 10 and 20 percent

Offer probability raised in the post-pandemic economy, psychological cost of therapy fixed.

	Fiscal effect (£1,000 per additional therapist)				Severe-depression
Workforce	Tax	Benefit savings	Therapist wages	Net	person-years averted
+5%	266	188	351	+103	557
+10%	226	146	316	+56	482
+20%	160	57	321	−103	437

Returns per therapist fall as the marginal course reaches a milder case: the net balance turns negative at 20%, though the health gain per therapist stays large.

Back

Additional counterfactuals

1. **Wage subsidy for poor mental health** ▸ Wage subsidy

2. **Reduce the stigma of attending therapy** ▸ Stigma

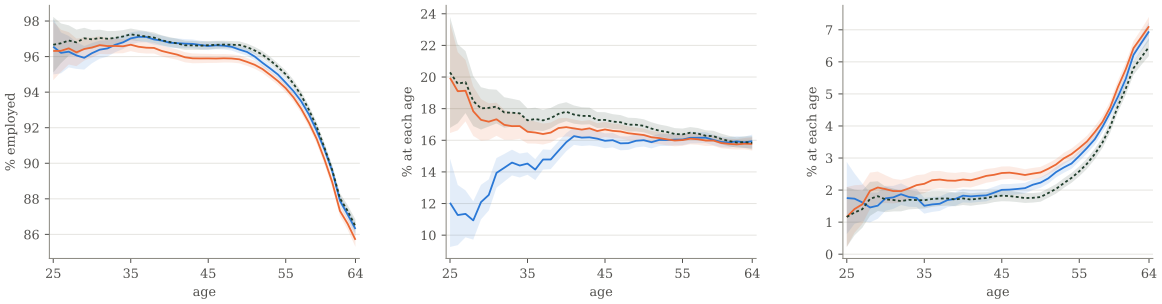
◂ Back

A wage subsidy raises work but worsens depression

→ Post-pandemic economy. The government pays the gap between each worker's offered wage and his wage in good mental health: a structural penalty of 8.3 log points at severe.

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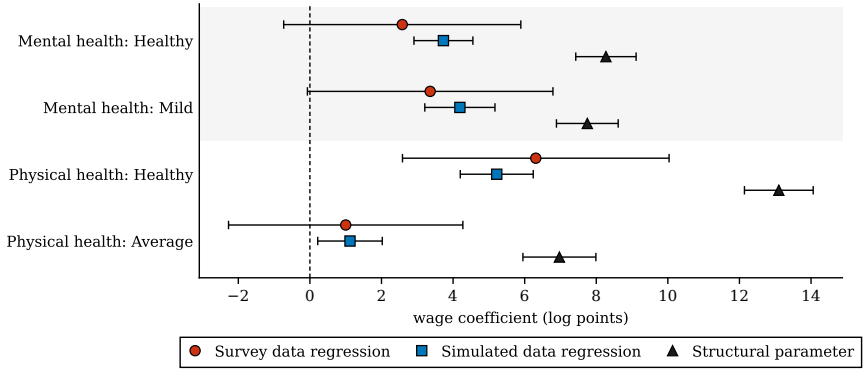
(a) Employment rate (b) Severe depression (c) Depression trap

Employment rises 0.73 pp and the trap shrinks, but severe depression rises: in work, seeking therapy is far more costly ($\psi(1) \gg \psi(0)$), so therapy use falls. Recovers ≈60% of its cost.

▸ Wage coefficients ◂ Back

Among workers, wage regressions understate the structural penalty

→ Log wages on mental and physical health (severe omitted) and low genetic risk, employed men: the survey regression, the same regression on simulated data, and the structural coefficients.



Mental health: 2.6 log points in the survey, indistinguishable from zero; 8.3 in the model. The men whose wages suffer most leave work, so wages among workers understate the penalty.

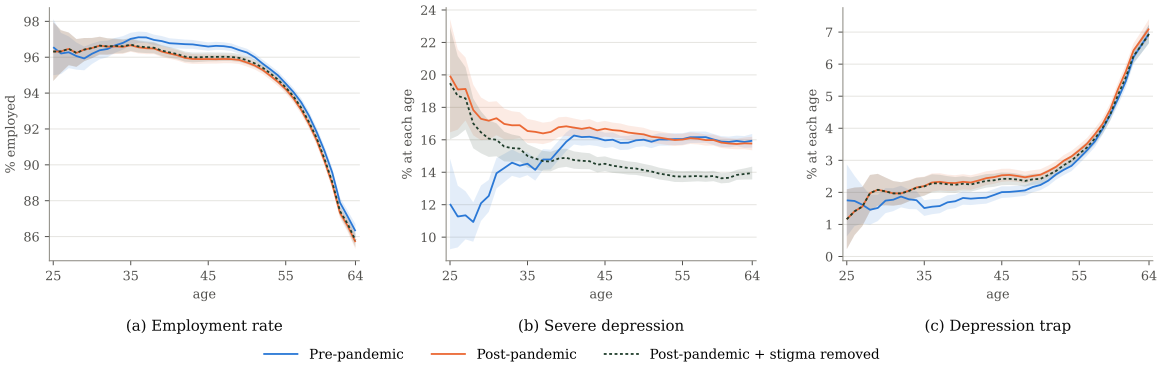
Back

Removing the cost of seeking therapy cuts severe depression most

$\rightarrow$ Post-pandemic economy. The estimated psychological cost of seeking therapy is about 75 times higher in work than out of it; here it is removed for both, $\psi(1) = \psi(0) = 0$.

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(a) Employment rate (b) Severe depression (c) Depression trap

Severe depression falls 2.06 pp, the most of any post-pandemic experiment, and from about age 40 lies below the pre-pandemic profile. It takes 30,600 more therapists to meet the demand.

▸ All reductions ◂ Back

Partial cuts to the psychological cost pay for the extra therapy

In-work psychological cost cut to the stated share of its estimate; out-of-work cost fixed except in the last row. Post-pandemic economy.

In-work cost retained	Severe depression	Employment rate	Courses per person	Additional therapists	Fiscal return per £1
100% (benchmark)	16.2%	93.0%	0.35	—	—
75%	−0.45 pp	+0.01 pp	0.61	7,400	1.14
50%	−0.96 pp	+0.02 pp	0.96	13,400	1.20
25%	−1.37 pp	+0.05 pp	1.32	17,800	1.16
0% (both costs)	−2.06 pp	+0.10 pp	2.44	30,600	0.96

The return per £1 of extra therapy peaks at halving the in-work cost; full removal falls just short. Excludes the cost of whatever lowers the psychological cost itself.

Back

Priority access: all priority levels

Capacity fixed at the benchmark; above-median-risk applicants wait X percent less than below-median. Every applicant is offered treatment with probability 0.69 in the benchmark.

	Treated (%), by risk		Fiscal effect (£ per person)			Severe-depression
Priority X	Below	Above	Tax	Benefits	Net	years averted per course
Benchmark	1.22	2.51	—	—	—	0.415
25%	1.03	2.72	4.8	7.9	12.8	0.444
50%	0.79	2.98	5.7	9.6	15.3	0.482
75%	0.45	3.33	8.5	16.8	25.3	0.534

Fiscal gains and the clinical return per course both rise with priority intensity; the treated share of below-median-risk men falls from 1.22% to 0.45%.

Back

Priority access: who applies for therapy

Therapy applicants, benchmark simulation. Priority moves courses from left to right.

	Below-median risk (lose access)	Above-median risk (gain access)
When they apply		
Severely depressed (%)	89	91
Employed (%)	68	69
Out of work and depressed (%)	20	22
Earnings if employed (£)	33,700	31,800
Poor physical health (%)	30	35
Who they are		
Aged 50 or over (%)	58	47
Previously treated (%)	60	49
Prior years severely depressed	4.0	2.4
Median assets (£)	22,400	12,300
Largest type (share, %)	Mostly depressed (77)	High earnings (63)

Equally sick and employed at application. Therapy moves from **older, chronic, repeat users** to **younger men earlier in their illness, with half the assets**.

Types Back

Genetic information: all priority levels

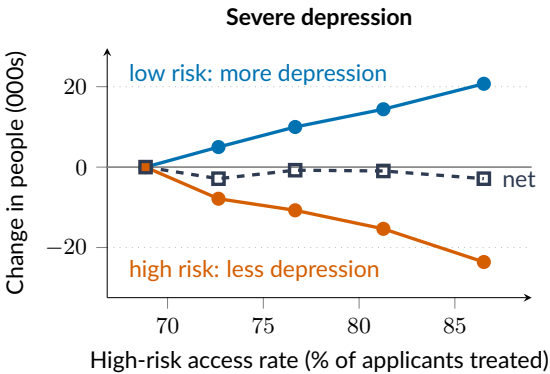
Without genotypes, a clinician must infer risk from health, work and earnings histories. In the survey those histories classify risk with an AUC of about 0.65.

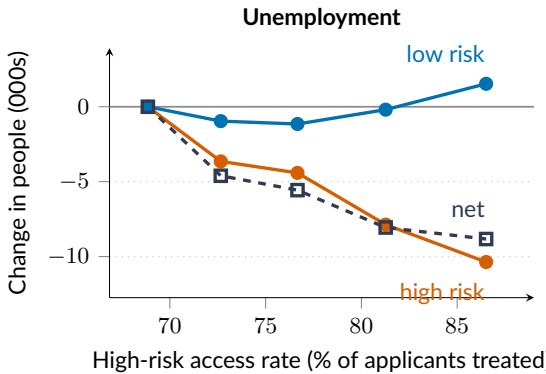
		Fiscal effect (£ per person)		Years of severe
Priority X	Information	Tax + benefits	Net	depression averted per course
Benchmark		—	—	0.415
25%	Genotype	4.8 + 7.9	12.8	0.444
	Histories only	0.6 + 0.1	0.7	0.421
50%	Genotype	5.7 + 9.6	15.3	0.482
	Histories only	2.7 + 4.1	6.8	0.429
75%	Genotype	8.5 + 16.8	25.3	0.534
	Histories only	2.7 + 4.6	7.4	0.437

Imperfect classification removes most of the fiscal and clinical gain at every priority level.

Back

Priority access: who gains and who loses

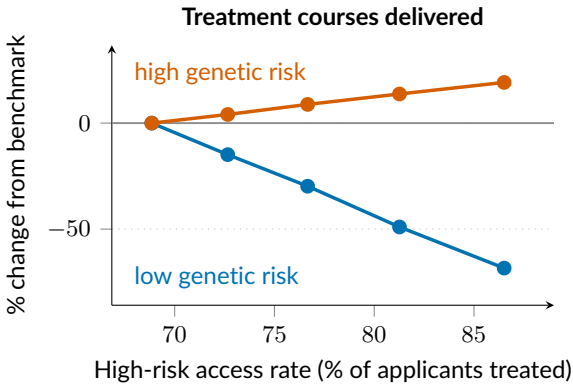




England scale, fixed capacity: severe depression $-24{,}000$ at high risk against $+21{,}000$ at low risk; employment $+10{,}400$ at high risk, net $+8{,}800$.

Volume Back

Priority access: how much treatment moves

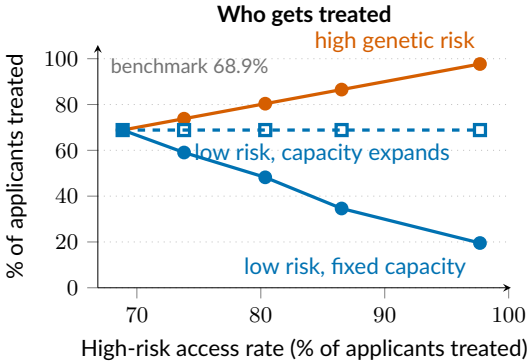


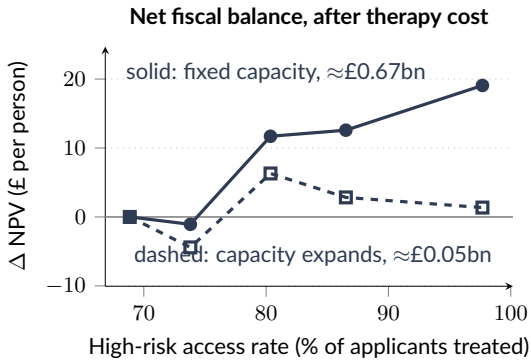
Capacity is fixed: every extra high-risk course is a low-risk course not delivered.

At the most intensive scenario, ≈190,000 of the 1.26m annual courses move: the low-risk group loses 68% of its courses and the high-risk group gains 19%.

Back

Priority access: pre-COVID initial distribution





Same sign, smaller return: £19.07 per person (≈£0.67bn) against £23.63 post-COVID, and the mildest scenario is slightly negative. The exchange rate is gentler, 1.9 against 2.9, as high risk takes 67% of volume here.

Post-COVID is *polarised* rather than uniformly worse; pre-COVID the gain comes mainly from *rationing* low-risk access.

Back

Redundancy: econometric specification

$$y_{it} = \sum_{\substack{k=-3 \\ k\neq -1}}^{+3} \delta_k \, \mathbf{1}\{t - t_0^i = k\} + \text{age}_{it} + \text{age}_{it}^2 + \underbrace{\nu_e}_{\text{education FE}} + \underbrace{\gamma_t}_{\text{year FE}} + \varepsilon_{it}$$

Estimator Sun & Abraham (2021) interaction-weighted DID. Reference $k = -1$; SEs clustered by individual.

Treatment (t_0^i) Earliest calendar year individual has a job ending due to redundancy (UKHLS & BHPS pooled).

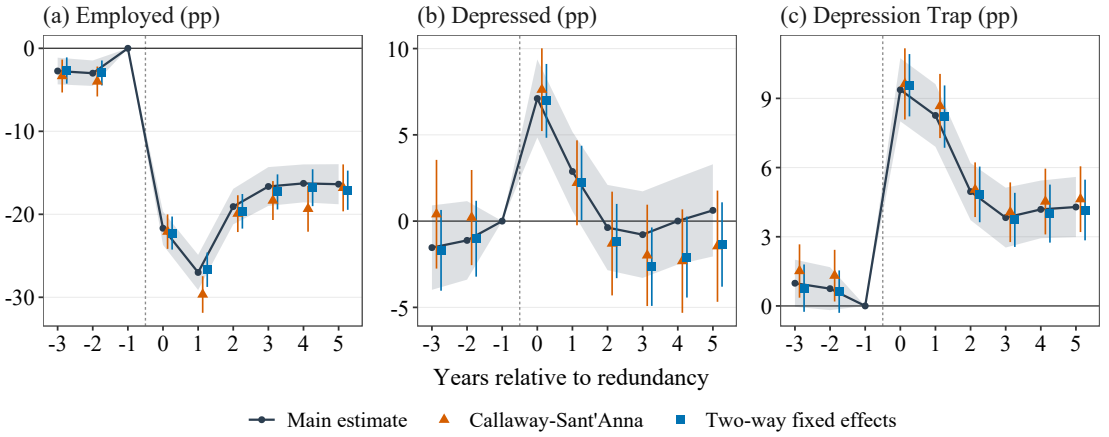
Sample Men aged 25–55, redundant vs. not-yet-redundant. **3,515** treated men in the ± 3 window (**591** balanced at all seven event times).

Outcomes GHQ-12 caseness count $(0\text{–}12)$; $\Pr(\text{employed})$.

G×E Median split of the depression PGI, residualised on the 10 genetic principal components.

Back

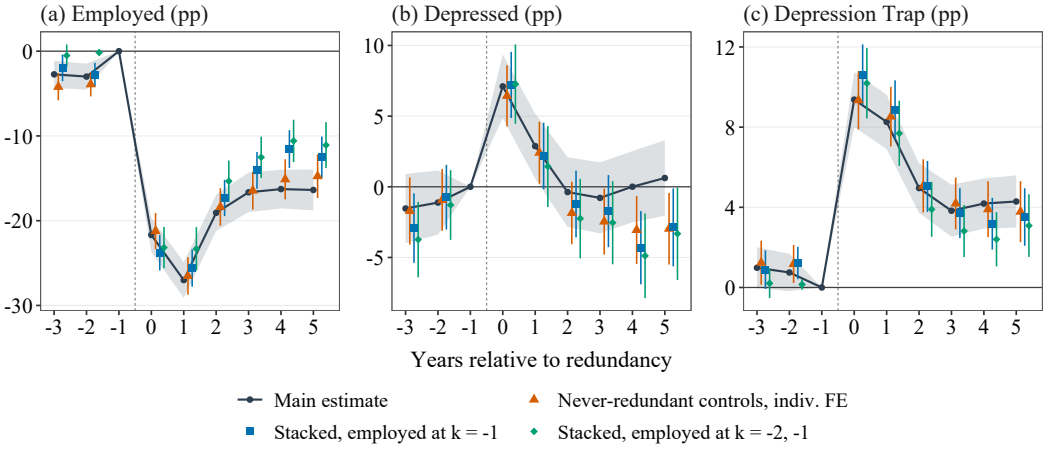
Other estimators give the same path



At impact (employed / depressed / trap, pp): Callaway–Sant'Anna, against men made redundant later, -22.1 / $+7.6$ / $+9.6$; two-way fixed effects -22.3 / $+7.0$ / $+9.6$; main -21.7 / $+7.1$ / $+9.4$.

Back

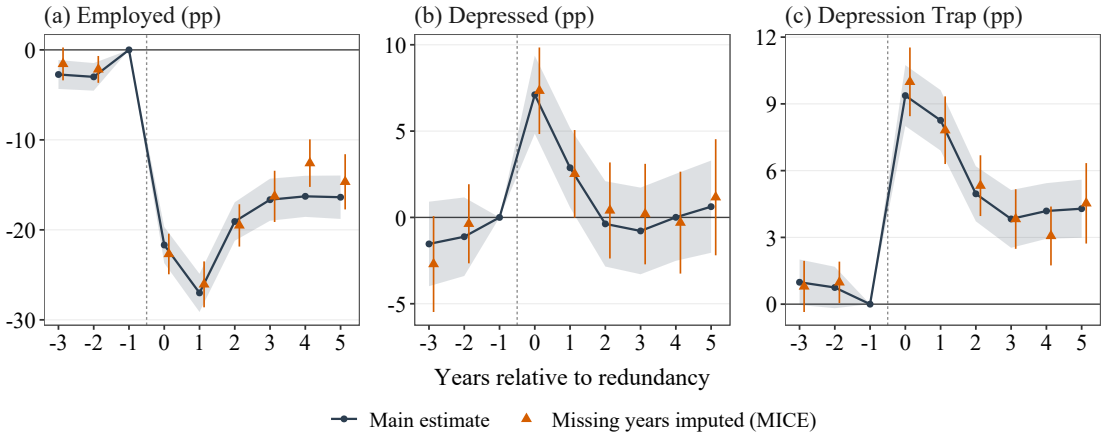
Other controls and individual fixed effects give the same impact



Individual FE with never-redundant men as controls, or stacked clean controls employed at $k = -1$ (or at -2 and -1). At impact (employed / depressed / trap, pp): -21.2 / $+6.4$ / $+9.3$; -23.8 / $+7.2$ / $+10.6$; -23.2 / $+7.3$ / $+10.2$.

Back

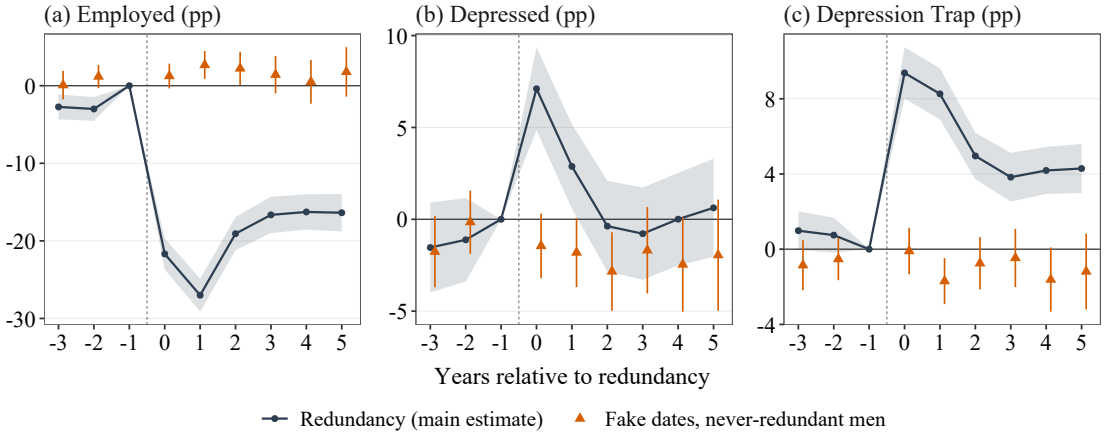
Imputing the years of men who leave the panel changes nothing



Panel exit rises after the event. For the 1,953 men seen at $k = -1$ and 0, missing years of employment and depression are imputed from pre-event levels, age and cohort (20 draws); the trap is built from them. At impact: -22.7 / $+7.3$ / $+10.0$ pp.

Back

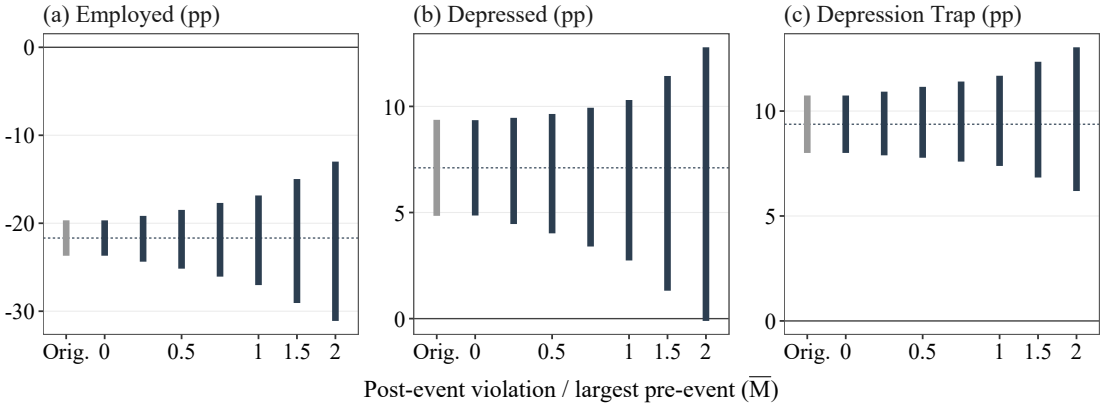
Placebo events produce no effect



Event dates drawn at random for men never made redundant, same design. At impact: employed $+1.3$ (s.e. 0.8), depressed -1.5 (0.9), trap -0.1 (0.6) pp.

Back

All three impacts survive plausible violations of parallel trends



Rambachan and Roth (2023) 95% sets for the impact at $k = 0$ (grey: conventional; dashed: estimate), allowing post-event violations up to $\bar{M}$ times the largest pre-event one. Employment and the trap exclude zero at $\bar{M} = 2$ ($[-31.1, -13.0]$; $[6.2, 13.0]$ pp); depression up to $\bar{M} = 1.5$ ($[1.3, 11.4]$).

Back

Recurrence: design and identification

Event A *recurrence*: a well $\rightarrow$ case crossing of GHQ-12 caseness after a recovered first episode (recovery = a sustained return below threshold).

Design Sun & Abraham (2021) around the recurrence wave; controls are *recovered, not-yet-recurred* men; calendar-wave FE, age and age^2; reference $k = -1$; SEs clustered by individual. Men 25–55, UKHLS & BHPS: **3,577** recurrences.

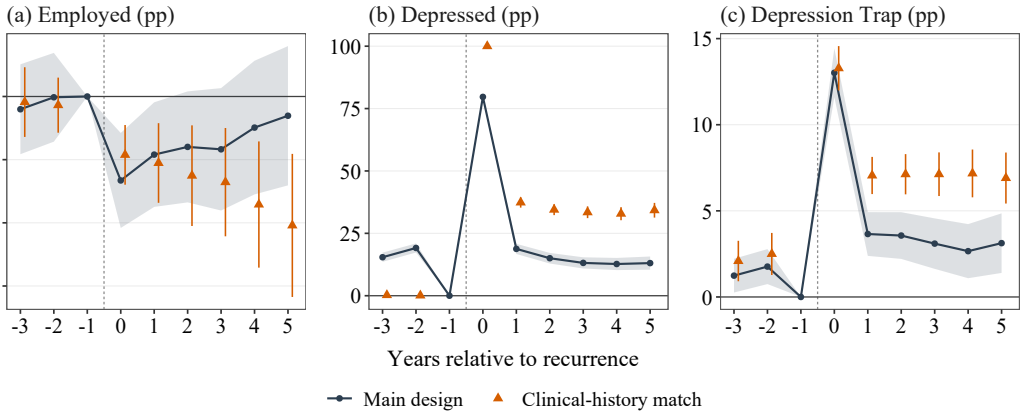
Assumption Conditional independence of recurrence *timing* given liability and clinical history — not literal randomness.

Why timing is plausibly quasi-random (kindling):

- Clinical evidence: first episodes are provoked, later episodes increasingly autonomous (Post, 1992; Kendler et al., 1999); conditional timing is largely unpredictable from clinical history.
- Within person, provoking events predict a **first onset far more than a recurrence**: separation/divorce raises first-onset risk by 28.1 pp against 3.0 for recurrence; redundancy 9.2 vs 4.9; a past-year redundancy does not predict a recurrence ($p = 0.61$).
- Robustness: the -2.7 pp headline is -2.6 excluding recurrences within a year of a redundancy; stable across estimators, comparison groups and caseness thresholds.

▸ Two Shocks ◂ Back

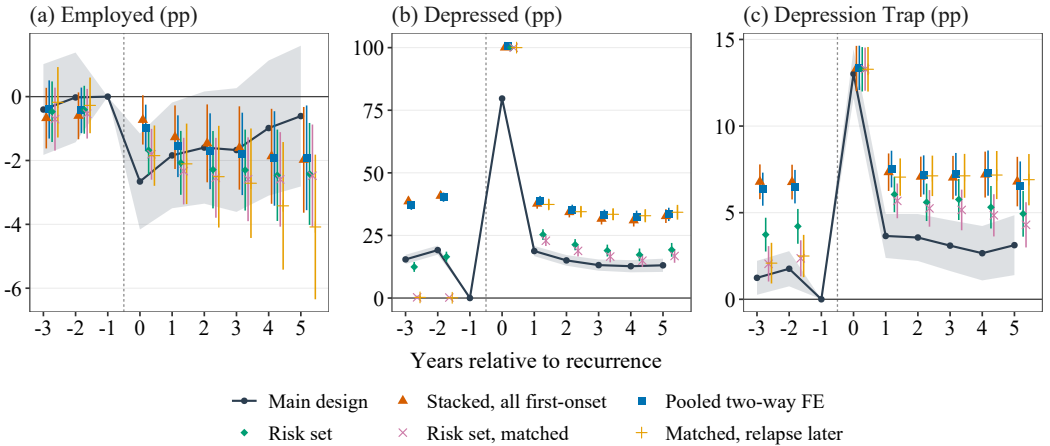
Same clinical history: employment pre-trends are flat



Line: the main slide's design. Points: controls with one earlier episode of the same length and time in remission, who relapse later (each dropped at his own relapse); individual FE. Employment (year up to each interview): -1.9 pp (s.e. 0.5) in the onset year. After $k = 0$ controls have not yet relapsed, so the depressed and trap gaps are the treated men's own levels.

Back

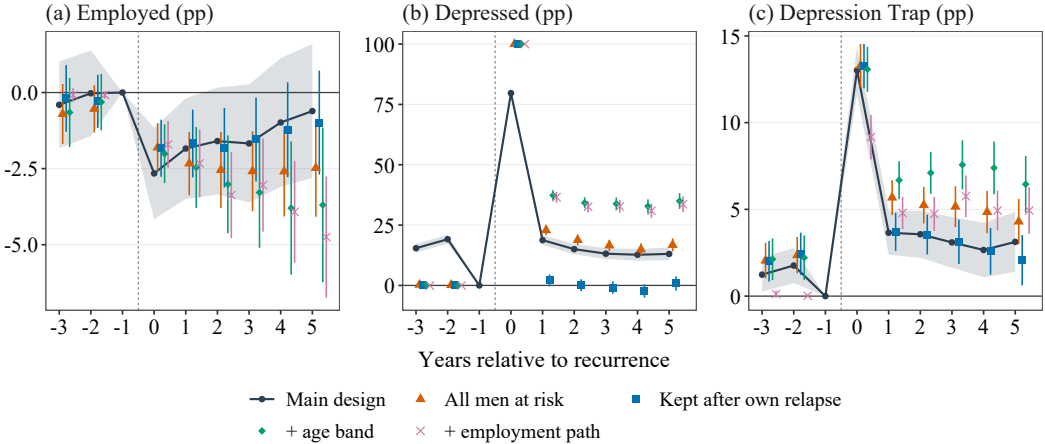
Fixed effects keep the loss; comparable controls remove the dip



Line: the main slide's design. Points: five designs with individual fixed effects. Against all first-onset men, depression and the trap are higher before the event, since a recurrence requires an earlier episode; matching on clinical history removes the depression gap. Employment in the onset year: -0.7 to -1.9 pp.

Back

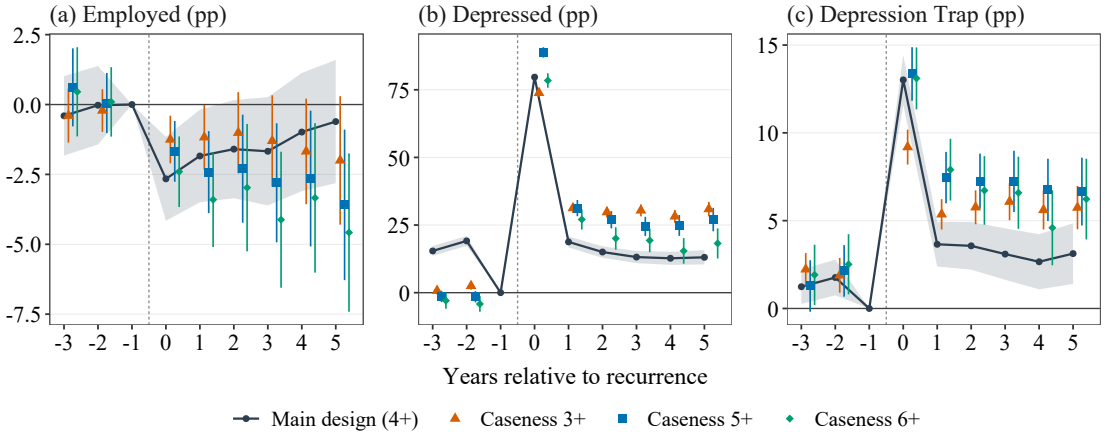
Other control groups and matches give the same impact



Line: the main slide's design. Points: all men at risk (-1.8 pp; trap $+5.7$ pp at $k = 1$); relapse-later controls kept after their own relapse (-1.8; trap $+3.7$); matched also on age band (-2.0) or on employment at $k = -3$ to -1 (-1.7; flat employment and trap before the event by construction).

Back

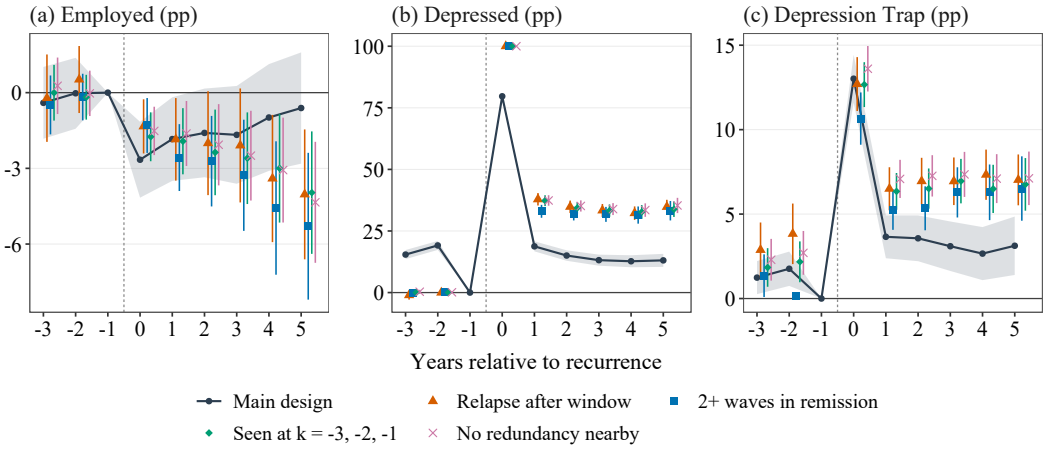
A more severe recurrence costs more work



Line: the main slide's design (caseness 4+). Points: the clinical-history design with the event re-dated at each cut; depressed is caseness 4+ throughout. In the year after the recurrence: employment -1.2 pp at 3+, -2.4 at 5+, -3.4 at 6+. Pre-event employment tests: $p = 0.63$ to 0.84.

Back

Recovery depth, attrition and job loss do not drive it



Line: the main slide's design. Points, in the clinical-history design: both groups two or more waves in remission (-1.3 pp, s.e. 0.6); men observed at $k = -3$, -2 and -1 (-1.8); recurrences within a year of a redundancy dropped (-1.5); controls relapsing after the window (-1.3).

Back

Arild Aakvik, Tor Helge Holmås, and Egil Kjerstad. Prioritization and the elusive effect on welfare – A Norwegian health care reform revisited. *Social Science & Medicine*, 128: 290–300, 2015. doi: 10.1016/j.socscimed.2015.01.044.

Boaz Abramson, Job Boerma, and Aleh Tsyvinski. Macroeconomics of mental health. Technical report, National Bureau of Economic Research, 2024.

David A. Adler, Thomas J. McLaughlin, William H. Rogers, Hong Chang, Leueen Lapitsky, and Debra Lerner. Job performance deficits due to depression. *American Journal of Psychiatry*, 163(9):1569–1576, 2006. doi: 10.1176/ajp.2006.163.9.1569.

Francesco Agostinelli and Zach Weingarten. Modeling and measuring the genetic determinants of child development. Working Paper 34427, National Bureau of Economic Research, October 2025. URL `http://www.nber.org/papers/w34427`.

George A. Akerlof. The economics of "tagging" as applied to the optimal income tax, welfare programs, and manpower planning. *American Economic Review*, 68(1):8–19, 1978.

Vikesh Amin, Jere R Behrman, Jason M Fletcher, Carlos A Flores, Alfonso Flores-Lagunes, and Hans-Peter Kohler. Mental health, schooling attainment and polygenic scores: Are there significant gene-environment associations? *bioRxiv*, page 684688, 2019.

Jehannine C Austin. Evidence-based genetic counseling for psychiatric disorders: a road map. *Cold Spring Harbor Perspectives in Medicine*, 10(6):a036608, 2020.

Eduardo Azevedo, Jonathan P. Beauchamp, and Richard Karlsson Linnér. Genetic prediction and adverse selection. *Econometrica*, 94(5):1817–1848, 2026. doi: 10.3982/ECTA24284.

Daniel Barth, Nicholas W Papageorge, and Kevin Thom. Genetic endowments and wealth inequality. *Journal of Political Economy*, 128(4):1474–1522, 2020.

Daniel Barth, Nicholas W Papageorge, Kevin Thom, and Mateo Velásquez-Giraldo. Genetic endowments, income dynamics, and wealth accumulation over the lifecycle. Technical report, National Bureau of Economic Research, 2022.

Bart ML Baselmans, Rick Jansen, Hill F Ip, Jenny van Dongen, Abdel Abdellaoui, Margot P van de Weijer, Yanchun Bao, Melissa Smart, Meena Kumari, Gonneke Willemsen, et al. Multivariate genome-wide analyses of the well-being spectrum. *Nature genetics*, 51(3): 445–451, 2019.

Daniel J. Benjamin, David Cesarini, Patrick Turley, and Alexander Strudwick Young. Social-science genomics: Progress, challenges, and future directions. NBER Working Paper 32404, National Bureau of Economic Research, 2024.

Barbara Biasi, Michael S. Dahl, and Petra Moser. Career effects of mental health. NBER Working Paper 29031, National Bureau of Economic Research, 2021.

Pietro Biroli, Titus J Galama, Stephanie von Hinke, Hans Van Kippersluis, Cornelius A Rietveld, and Kevin Thom. The economics and econometrics of gene-environment interplay. *arXiv preprint arXiv:2203.00729*, 2022.

Uta Bolt, Eric French, Varun Warrier, Qianyu Yang, and Weilong Zhang. Genetic endowments and lifetime earnings: Understanding the mechanisms. 2025.

Stéphane Bonhomme, Thibaut Lamadon, and Elena Manresa. Discretizing unobserved heterogeneity. *Econometrica*, 90(2):625–643, 2022.

Mark L. Bryan, Nigel Rice, Jennifer Roberts, and Cristina Sechel. Mental health and employment: A bounding approach using panel data. *Oxford Bulletin of Economics and Statistics*, 84(5):1018–1051, 2022.

Elena Capatina and Michael Keane. Health shocks, health insurance, human capital, and the dynamics of earnings and health. *Journal of Political Economy*, 134(6):1714–1774, 2026.

Andrea Cipriani, Toshi A. Furukawa, Georgia Salanti, Anna Chaimani, Lauren Z. Atkinson, Yusuke Ogawa, Stefan Leucht, Henricus G. Ruhe, Erick H. Turner, Julian P. T. Higgins, Matthias Egger, Nozomi Takeshima, Yu Hayasaka, Hissei Imai, Kiyomi Shinohara, Aran Tajika, John P. A. Ioannidis, and John R. Geddes. Comparative efficacy and acceptability of 21 antidepressant drugs for the acute treatment of adults with major depressive

disorder: a systematic review and network meta-analysis. *The Lancet*, 391(10128): 1357–1366, 2018. doi: 10.1016/S0140-6736(17)32802-7. URL https://doi.org/10.1016/S0140-6736(17)32802-7.

Sheldon Cohen and Thomas A. Wills. Stress, social support, and the buffering hypothesis. *Psychological Bulletin*, 98(2):310–357, 1985. doi: 10.1037/0033-2909.98.2.310.

Christopher J Cronin, Matthew P Forsstrom, and Nicholas W Papageorge. What good are treatment effects without treatment? mental health and the reluctance to use talk therapy. *Review of Economic Studies*, 2024.

Pim Cuijpers, Annemieke van Straten, and Lisanne Warmerdam. Behavioral activation treatments of depression: A meta-analysis. *Clinical Psychology Review*, 27(3):318–326, 2007. doi: 10.1016/j.cpr.2006.11.001.

Mariacristina De Nardi, Eric French, and John Bailey Jones. Medicaid insurance in old age. *American Economic Review*, 106(11):3480–3520, 2016. doi: 10.1257/aer.20140015.

Mariacristina De Nardi, Svetlana Pashchenko, and Ponpoje Porapakkarm. The lifetime costs of bad health. *Review of Economic Studies*, 2024.

Jonathan De Quidt and Johannes Haushofer. Depression for economists. Technical report, National Bureau of Economic Research, 2016.

Department of Health and Social Care. Fit for the future: 10 year health plan for England. Technical report, HM Government, London, 2025. URL https://www.gov.uk/government/publications/10-year-health-plan-for-england-fit-for-the-future. Published 3 July 2025.

Cecilia S. Diaz-Campo, Barton H. Hamilton, Enrique Martin Luccioni, and Hyun Soo Suh. Mental health and the early career dynamics of young men. SSRN Working Paper 5429296, November 2025. URL https://ssrn.com/abstract=5429296.

Joe Dodd, Luke Munford, Matt Sutton, and Igor Francetic. The effect of area-level waiting times for psychological therapies on individual-level labour market outcomes. *Labour Economics*, 96:102748, 2025. doi: 10.1016/j.labeco.2025.102748.

Mark M. Drozd. Mental health, educational attainment, and occupational choice. Job Market Paper, Johns Hopkins University, June 2026.

Susan L Ettner, Richard G Frank, and Ronald C Kessler. The impact of psychiatric disorders on labor market outcomes. *ILR Review*, 51(1):64–81, 1997.

Ellie M Fossey and Carol A Harvey. Finding and sustaining employment: a qualitative meta-synthesis of mental health consumer views. *Canadian Journal of Occupational Therapy*, 77(5):303–314, 2010.

Richard G. Frank and Thomas G. McGuire. Economics and mental health. In Anthony J. Culyer and Joseph P. Newhouse, editors, *Handbook of Health Economics*, volume 1B, pages 893–954. Elsevier, 2000. doi: 10.1016/S1574-0064(00)80029-3.

Eric French. The effects of health, wealth, and wages on labour supply and retirement behaviour. *The Review of Economic Studies*, 72(2):395–427, 2005.

Eric French and Jae Song. The effect of disability insurance receipt on labor supply. *American Economic Journal: Economic Policy*, 6(2):291–337, 2014. doi: 10.1257/pol.6.2.291.

Paul Frijters, David W Johnston, and Michael A Shields. The effect of mental health on employment: evidence from australian panel data. *Health economics*, 23(9):1058–1071, 2014.

Anna Godøy, Venke F. Haaland, Ingrid Huitfeldt, and Mark Votruba. Hospital queues, patient health, and labor supply. *American Economic Journal: Economic Policy*, 16(2): 150–181, 2024. doi: 10.1257/pol.20210399.

Hugh Gravelle and Luigi Siciliani. Third degree waiting time discrimination: Optimal allocation of a public sector healthcare treatment under rationing by waiting. *Health Economics*, 18(8):977–986, 2009. doi: 10.1002/hec.1423.

Nathaniel Hendren and Ben Sprung-Keyser. A unified welfare analysis of government policies. *Quarterly Journal of Economics*, 135(3):1209–1318, 2020. doi: 10.1093/qje/qjaa006.

Stefan G. Hofmann, Anu Asnaani, Imke J. J. Vonk, Alice T. Sawyer, and Angela Fang. The efficacy of cognitive behavioral therapy: A review of meta-analyses. *Cognitive Therapy and Research*, 36(5):427–440, 2012. doi: 10.1007/s10608-012-9476-1. URL https://doi.org/10.1007/s10608-012-9476-1.

Roozbeh Hosseini, Karen A Kopecky, and Kai Zhao. How important is health inequality for lifetime earnings inequality? Technical report, Working paper, 2021.

Mikkel Aagaard Houmark, Victor Ronda, and Michael Rosholm. The nurture of nature and the nature of nurture: How genes and investments interact in the formation of skills. *American Economic Review*, 114(2):385–425, 2024.

Reshmaan Hussam, Erin M. Kelley, Lane Gregory, and Fatima Zahra. The psychosocial value of employment: Evidence from a refugee camp. *American Economic Review*, 112 (11):3691–3724, 2022. doi: 10.1257/aer.20211517.

Marie Jahoda. *Employment and Unemployment: A Social-Psychological Analysis*. Cambridge University Press, Cambridge, 1982.

Grégory Jolivet and Fabien Postel-Vinay. A structural analysis of mental health and labour market trajectories. *Review of Economic Studies*, 92(3):1920–1954, 2025.

Kenneth S. Kendler, Laura M. Karkowski, and Carol A. Prescott. Causal relationship between stressful life events and the onset of major depression. *American Journal of Psychiatry*, 156(6):837–841, 1999.

Toru Kitagawa and Aleksey Tetenov. Who should be treated? Empirical welfare maximization methods for treatment choice. *Econometrica*, 86(2):591–616, 2018. doi: 10.3982/ECTA13288.

Jon Kleinberg, Jens Ludwig, Sendhil Mullainathan, and Ziad Obermeyer. Prediction policy problems. *American Economic Review*, 105(5):491–495, 2015. doi: 10.1257/aer.p20151023.

Marit Knapstad, Linn V. Lervik, Silje M. M. Sæther, Leif E. Aarø, and Otto R. F. Smith. Effectiveness of prompt mental health care, the Norwegian version of improving access to psychological therapies: A randomized controlled trial. *Psychotherapy and Psychosomatics*, 89(2):90–105, 2020.

Andreas Kuhn, Rafael Lalive, and Josef Zweimüller. The public health costs of job loss. *Journal of Health Economics*, 28(6):1099–1115, 2009.

Richard Layard, David Clark, Martin Knapp, and Guy Mayraz. Cost-benefit analysis of psychological therapy. *National Institute Economic Review*, 202:90–98, 2007. doi: 10.1177/0027950107086171.

Matthew S Lebowitz and Woo-kyoung Ahn. Blue genes? understanding and mitigating negative consequences of personalized information about genetic risk for depression. *Journal of genetic counseling*, 27(1):204–216, 2018.

Cotton M. Lindsay and Bernard Feigenbaum. Rationing by waiting lists. *American Economic Review*, 74(3):404–417, 1984.

Kyle Logue and Joel Slemrod. Genes as tags: The tax implications of widely available genetic information. *National Tax Journal*, 61(4):843–863, 2008. doi: 10.17310/ntj.2008.4S.04.

Hamish Low and Luigi Pistaferri. Disability insurance and the dynamics of the incentive insurance trade-off. *American Economic Review*, 105(10):2986–3029, 2015. doi: 10.1257/aer.20110108.

James Hentall Maccuish. Costly attention and retirement. Technical report, Working paper, University College of London, 2022.

N. Gregory Mankiw and Matthew Weinzierl. The optimal taxation of height: A case study of utilitarian income redistribution. *American Economic Journal: Economic Policy*, 2(1): 155–176, 2010. doi: 10.1257/pol.2.1.155.

Jan Marcus. The effect of unemployment on the mental health of spouses – evidence from plant closures in Germany. *Journal of Health Economics*, 32(3):546–558, 2013.

Nicolau Martin Bassols, Sonja De New, John P. De New, Clement Wong, and Danusha Jayawardana. The impact of mental health on labour market outcomes: New evidence from genetic markers. Working paper, 2024.

David McDaid and A-La Park. The economic case for investing in the prevention of mental health conditions in the UK. Technical report, Mental Health Foundation and London School of Economics and Political Science, London, 2022.

Andrew S. Moriarty, Nick Meader, Kym I. E. Snell, Richard D. Riley, Lewis W. Paton, Carolyn A. Chew-Graham, Simon Gilbody, Rachel Churchill, Robert S. Phillips, Shehzad Ali, and Dean McMillan. Prognostic models for predicting relapse or recurrence of major depressive disorder in adults. *Cochrane Database of Systematic Reviews*, (5):CD013491, 2021.

Sarah Morris, Suzanne Hill, Sally McManus, and Traolach Brugha. Adult psychiatric morbidity survey: Survey of mental health and wellbeing, England, 2023/4. Technical

report, National Centre for Social Research, for NHS England, London, November 2025. URL https://natcen.ac.uk/publications/adult-psychiatric-morbidity-survey-20234.

NHS England. 2021/22 national cost collection data publication: National schedule of NHS costs. https://www.england.nhs.uk/publication/2021-22-national-cost-collection-data-publication/, 2023a. Published 5 April 2023. PLICS IAPT care contacts: national average unit cost £125.78 per attended contact (NHS trusts and foundation trusts).

NHS England. NHS Long Term Workforce Plan. Technical report, NHS England, London, UK, June 2023b.

NHS England. NHS Talking Therapies for anxiety and depression, Annual Report 2022–23. Annual report, NHS England Digital, Leeds, 2023c. URL https://digital.nhs.uk/data-and-information/publications/statistical/nhs-talking-therapies-for-anxiety-and-depression-annual-reports/2022-23. Referrals completing a course had 8.1 treatment sessions on average, up from 7.9 in 2021-22.

NHS England. Mental health bulletin, 2022-23 annual report. https://digital.nhs.uk/data-and-information/publications/statistical/

mental-health-bulletin/2022-23-annual-report, 2024. Published 22 February 2024. 3,582,864 people in contact with secondary mental health, learning disability and autism services.

NHS England. Medium term planning framework 2026/27 to 2028/29. Technical report, NHS England, London, UK, 2025. Talking-therapies course target: 805,000 in 2026/27 to 915,000 in 2028/29.

Susan Nolen-Hoeksema, Blair E. Wisco, and Sonja Lyubomirsky. Rethinking rumination. *Perspectives on Psychological Science*, 3(5):400–424, 2008. doi: 10.1111/j.1745-6924.2008.00088.x.

Office for National Statistics. Rising ill-health and economic inactivity because of long-term sickness, UK: 2019 to 2023. https://www.ons.gov.uk/employmentandlabourmarket/peoplenotinwork/economicinactivity/articles/risingillhealthandeconomicinactivitybecauseoflongtermsicknessuk/2019to2023, 2023. Released 26 July 2023.

Office for National Statistics. LFS: Economic inactivity by reason: long-term sick, UK, aged 16 to 64, seasonally adjusted (series LF69). Labour market statistics time series, released 18 August 2026, 2026. URL https://www.ons.gov.uk/employmentandlabourmarket/

peoplenotinwork/economicinactivity/timeseries/lf69/lms. 2.77 million in March to May 2026; peak of 2.85 million in August to October 2023. Accessed 11 September 2026.

Aysu Okbay, Yeda Wu, Nancy Wang, Hariharan Jayashankar, Michael Bennett, Seyed Moeen Nehzati, Julia Sidorenko, Hyeokmoon Kweon, Grant Goldman, Tamara Gjorgjieva, et al. Polygenic prediction of educational attainment within and between families from genome-wide association analyses in 3 million individuals. *Nature genetics*, 54(4):437–449, 2022.

Ekaterina Oparina, Christian Krekel, and Sorawoot Srisuma. Talking therapy: Impacts of a nationwide mental health service in england. *IZA Discussion Paper No. 16839*, 2024.

Nicholas W Papageorge and Kevin Thom. Genes, education, and labor market outcomes: evidence from the health and retirement study. *Journal of the European Economic Association*, 18(3):1351–1399, 2020.

Karsten I Paul and Klaus Moser. Unemployment impairs mental health: Meta-analyses. *Journal of Vocational behavior*, 74(3):264–282, 2009.

Matteo Picchio and Michele Ubaldi. Unemployment and health: A meta-analysis. *Journal of Economic Surveys*, 38(4):1437–1472, 2024.

Robert M. Post. Transduction of psychosocial stress into the neurobiology of recurrent affective disorder. *American Journal of Psychiatry*, 149(8):999–1010, 1992.

Roger Prudon. Is delayed mental health treatment detrimental to employment? *The Review of Economics and Statistics*, 2026. Forthcoming.

Timothy G. Raben, Louis Lello, Erik Widen, and Stephen D. H. Hsu. From genotype to phenotype: Polygenic prediction of complex human traits. In Nourollah Ahmadi and Jérôme Bartholomé, editors, *Genomic Prediction of Complex Traits: Methods and Protocols*, volume 2467 of *Methods in Molecular Biology*, pages 421–446. Springer, New York, NY, 2022. doi: 10.1007/978-1-0716-2205-6_15.

Ashesh Rambachan and Jonathan Roth. A more credible approach to parallel trends. *The Review of Economic Studies*, 90(5):2555–2591, 2023.

Matthew Ridley, Gautam Rao, Frank Schilbach, and Vikram Patel. Poverty, depression, and anxiety: Causal evidence and mechanisms. *Science*, 370(6522):eaay0214, 2020.

Robert M Sauer, Noemi Mantovan, and Guido Cozzi. How do mental and physical health influence career and family choices? Technical report, Centre for Household, Income, Labour and Demographic Economics (CHILD)-CCA, 2025.

Benjamin Ly Serena. The causal effect of scaling up access to psychotherapy. *Review of Economics and Statistics*, 2025. doi: 10.1162/rest.a.1679. Published online 18 December 2025.

Otto RF Smith, David M Clark, Gunnel Hensing, Richard Layard, and Marit Knapstad. Cost–benefit of iapt norway and effects on work-related outcomes and health care utilization: results from a randomized controlled trial using registry-based data. *Psychological Medicine*, 55:e86, 2025.

Tom Sterud, Lars-Kristian Lunde, Rigmor Berg, Karin I Proper, and Fiona Aanesen. Mental health effects of unemployment and re-employment: a systematic review and meta-analysis of longitudinal studies. *Occupational and environmental medicine*, 82(7): 343–353, 2025.

Liyang Sun and Sarah Abraham. Estimating dynamic treatment effects in event studies with heterogeneous treatment effects. *Journal of Econometrics*, 225(2):175–199, 2021.

Philip S. Wang, Patricia Berglund, Mark Olfson, Harold A. Pincus, Kenneth B. Wells, and Ronald C. Kessler. Failure and delay in initial treatment contact after first onset of mental disorders in the National Comorbidity Survey Replication. *Archives of General Psychiatry*, 62(6):603–613, 2005. doi: 10.1001/archpsyc.62.6.603.

Qianyu Yang, Naijia Guo, and Weilong Zhang. The DNA of enterprise: Genetic endowments, entrepreneurship and intergenerational mobility. Cambridge Working Papers in Economics 2668, Faculty of Economics, University of Cambridge, 2026.